

GE Healthcare

MSK Extreme Installation Guide



DOC0922201
Rev 3.0
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Revision History

Revision	Date	Changes
1.0	May, 2012	Initial conversion of 2000-9305 Rev E to GE format
2	December, 2012	Updated Chapter 2 with information for the AMETEK XG6-220 power supply. Updated Table B-4.
3	July, 2014	Updated the following sections in Chapter 2: Energizing (Ramping) the Magnet Using the XFR6-200 Power Supply Energizing (Ramping) the Magnet Using the XG6-220 Power Supply Ramping Down the Magnet Using the XG6-220 Power Supply The information added to this Install guide does not need to be validated. No new procedures were added and the changes made were reviewed and approved in MWS. The signatures in DOC0591190 are archived in the service validation binder.

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1 Overview

This guide explains how to install the GE Healthcare MR system at a customer site. The guide applies to model AB 2000 systems. The information contained in this document is the property of GE Healthcare. It may not be copied, printed, or duplicated by any means, without written permission from GE Healthcare.

Contents

The Installation Guide contains the following chapters:

- Chapter 1, [Overview](#) is this chapter. It lists the major sections of the Installation guide, and explains the text conventions used.
- Chapter 2, [System Installation](#) explains how to install the system hardware.
- Chapter 3, [System Configuration and Test](#) explains how to configure the system and test its functions.
- Appendix A, [Reference Tables](#) contains tables of connector pinouts, resistance-to-temperature conversions, and level-to-volume conversions.
- Appendix B, [Shipping Box Contents and Installation Tool Kits](#) lists the contents of the service installation tool kits and the system shipping containers.

Related Documents

Much of the information necessary for installing the GE Healthcare MR system is found on the *System Installation Data Sheet* (DOC0922205).

The *Site Planning Guide* (DOC0922291) explains in detail what features the site must have before installation begins.

When problems with equipment setup arise, the solutions can often be found in equipment service manuals on the Service CD. For instance, if the oil mist filter in the pump kit contains excessive oil, the mist filter service manual explains how to remove the oil.

Navigation

Links

[Blue text](#) indicates a link to another document, or to another section of the *Installation Guide*. Part numbers in blue are links to the related line in the parts list. Clicking the blue text jumps to the indicated place. (The blue text at the beginning of this item is not a link.)

All listings in the Table of Contents and in the Index are links to the named item (these links are not blue). Clicking a listing in the Table of Contents or the Index jumps to the listed page.

Bookmarks

The Bookmarks tab at the top left corner of the Acrobat screen provides a hierarchy of headings in the document that works like the Windows explorer. Clicking a heading in the bookmarks list opens the page that contains the heading. To open or close the bookmarks display, click the Bookmarks tab. Closing the Bookmarks display allows a larger display area for the document.

Switching Between Documents

After making a jump to another PDF document or to another location in the *Installation Guide*, use the back (left arrow) button at the bottom of the Adobe Reader window to jump back to the original location.

After jumping to a non-PDF document, press Alt-Tab to return to this document. (See [step 3](#) through [step 5](#) on page 50 for a detailed example of using Alt-Tab to switch between documents.)

Figure 1-1: Adobe Reader Back Button



Conventions

The conventions described below are used throughout the GE Healthcare MR service documents.

Text Conventions

The following text conventions indicate on-screen features and controls, and emphasize important operations:

- References to on-screen text and controls are in a special font, to distinguish them from the normal flow of text.

Example: When Clone completed Successfully appears, remove the floppy and CD-ROM disks.

In the example, “Clone completed Successfully” is text in the computer display.

- When a reference to on-screen text includes an undefined variable, the variable is in italics, between brackets.

Example: No connection to Server <Orthone:[*database number*]>

In the example, “[*database number*]” refers to a value that is not specified. The rest of the example is the same as the display text.

- In some procedures, the most significant words are in **bold type**. If the words refer to an on-screen feature, they are in the **special font in bold**.

Example: On the Start Copying Files window, click Next>.

In the example, the N in “Next” is underlined because it is underlined in the display.

Terminology

The following terms have the meanings listed here:

- “Press” means to press a key or switch on a keyboard or other input device. (It does not refer to operating on-screen controls, so the name of the control is in the regular font.)

Example: Press **Enter**.

- “Click” means to place the mouse cursor on an item on the screen, and press the left mouse button.

Example: Click Finish.

Clicking causes the computer to perform an operation, such as opening or closing a window, saving data, or executing a task.

- “Double-click” means to place the mouse cursor on an item on the screen, and press the left mouse button twice.

Example: Double-click the Network icon.

- “Right-click” means to place the mouse cursor on an item on the screen, and press the right mouse button.

Example: Right-click Onibc and click Open.

- “Select” means to pick an option or file name from a list, by clicking it or rolling the mouse cursor over it.

Example: From the Edit menu, select Copy.

Selecting usually has no immediate result, and requires a subsequent click operation to implement the selection.

Warnings, Cautions, and Notes

WARNING: Warnings describe actions or conditions that can cause injury or death. Never perform an action or create a condition described in a warning.

Caution: Cautions describe actions or conditions that can damage equipment. Never perform an action or create a condition described in a caution.

Note: Notes contain important information that can help you perform service operations. A note can describe a special condition you may encounter, can describe actions that simplify your work, or can alert you to other actions that create unnecessary work.

Safety Considerations

For detailed safety information relating to the ONI MR imaging system, refer to the *Safety Guide*. The following warnings apply to procedures in the *Installation Guide*:

WARNING: *Pressurized gas bottles can be dangerous if they fall over. Always make sure that gas bottles are secured to a wall or a stable cart, or place the bottles horizontally on the floor.*

WARNING: *Boil-off from cryogenic liquids can displace the breathable air in a confined space. To prevent possibly fatal asphyxiation, ensure that there is adequate ventilation, that the space is not confined, and that you do not remain in the area longer than necessary.*

WARNING: *Potential cold burn hazard! You may be exposed to cryogenics or contact cold objects during service procedures or a magnet quench. When performing any service procedure that uses cryogenics, wear protective clothing:*

- Long sleeve shirt
- Long pants
- Protective apron or jacket
- Non-absorbent gloves
- Glasses or goggles and a face shield

WARNING: *The ONI MR system creates a strong magnetic field, which can turn metal objects into missiles. To prevent personal injury and equipment damage, thoroughly search the exam room, and remove all tools and other objects that might be affected by the magnetic field, before energizing the magnet.*

Recommended PPE

- Safety glasses
- Face shield
- Cryogenic gloves
- Work gloves
- Cryogenic coat
- LOTO devices
- Oxygen monitor
- Non-ferrous safety shoes

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2 System Installation

Chapter Contents

This chapter includes instructions for installing an GE MR imaging system magnet at a customer site. The chapter includes the following topics:

- [Materials Required](#) on page 10
- [Unpacking the Electronics Cabinet and Patient Chair](#) on page 11
- [Installing the Magnet Dolly](#) on page 12
- [Placing the Magnet](#) on page 14
- [Aligning the Coldhead](#) on page 15
- [Installing the Coldhead](#) on page 18
- [Pumping out the Magnet](#) on page 18
- [Equipment Installation](#) on page 24
- [Connecting the Cable Harness](#) on page 27
- [Removing the Pump Assembly](#) on page 28
- [Powering up the System](#) on page 28
- [Cooling Down the Magnet](#) on page 31
- [Ramping the Magnet](#) on page 42
- [Ramping Down the Magnet Using the XFR6-200 Power Supply](#) on page 47
- [Ramping Down the Magnet Using the XG6-220 Power Supply](#) on page 48

Materials Required

- Magnet dolly tool kit (see [page 84](#))
- Vacuum pump tool kit (see [page 85](#))
- Helium transfer tool kit (see [page 83](#))
- Ramp tool kit (see [page 84](#))
- Coldhead plate alignment tool (see [page 17](#))
- Service CD P/ N 2000-9300-001
- Power screwdriver
- Standard tools
- Non-magnetic tools
- Feeler gauge or similar
- Precision caliper
- Anti-seize compound
- Vacuum grease
- Bottle of pure helium gas (five 9s purity)
- Dewar of liquid nitrogen
- Dewar of liquid helium

WARNING: *Pressurized gas bottles can be dangerous if they fall over. Always make sure that gas bottles are secured to a wall or a stable cart, or place the bottles horizontally on the floor.*

Note: All O-rings in the GE Healthcare MR system and in the pumping equipment must be lubricated with vacuum grease to prevent leaks.

Unpacking the Electronics Cabinet and Patient Chair

Figure 2-2: Shipping Crate Ramp

1. Remove the screws that secure the top panel and the two side panels of the crate to the ramp.
2. Remove the ramp from the shipping crate, and place it on the end of the crate base.
3. Carefully roll the electronics cabinet or patient chair down the ramp to its operating location.

Note: Do not roll the patient chair across very dirty or coarse floors or the casters may be damaged.



Unpacking the Magnet

The following procedure describes how to properly uncrate the magnet.

1. Closely inspect the shock indicators on the box (see Figure 2-1), inside the box, and on the magnet itself. If any indicators are activated, make a note and continue to unpack the magnet. Report any damage found.
2. Remove the screws that secure the top of the crate.
3. Lift off the top of the crate.
4. Inspect the magnet to verify that it is properly supported by the shipping cradle.

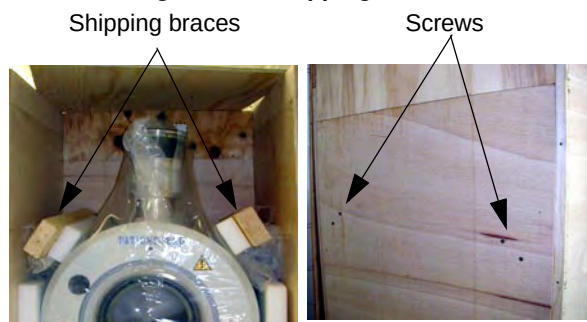
Figure 2-1: Shock Indicator



Caution: If the magnet is resting against the side of the crate, *do not continue uncrating it!* Arrange for a rigger to remove the magnet from the crate.

5. Four **screws** on each end of the crate secure the braces inside the crate. Remove the four screws from each end (see Figure 2-2).
There are two screws near the right edge, and two near the left edge, on each end of the crate. (See the figure at far right.)
6. Remove the screws that secure the four sides of the crate, and remove the sides.
7. Remove the plastic wrap from the magnet and inspect the magnet for damage.
8. Either have the movers transport the magnet to the exam room before you install the magnet dolly, or install the magnet dolly on the dock and use that to transport the magnet.

Figure 2-2: Shipping Braces and Screws



Installing the Magnet Dolly

All the parts required to install the magnet dolly are included in the kit (see [Magnet Dolly Tool Kit](#) on page 84.)

Figure 2-3: Washers on Captive Screws

1. Open the magnet dolly tool kit (See [page 84](#)), and remove its contents.
2. Place four washers on each captive screw (see [Figure 2-3](#)), and attach the **side plates** to the magnet with the taped side of each plate facing the magnet. Do not tighten the captive screws.



Figure 2-4: Side Plate Orientation

Note: The slotted holes in the plates must be toward the service end (footrest end) of the magnet.



Figure 2-5: Mounting End Plates on Side Plates

3. Secure the dolly end plates to the side plates (see [Figure 2-5](#)), with M8x50 mm hex bolts. Install lock washers and flat washers on the top and bottom bolts at each corner of the dolly, but not on the middle bolt.

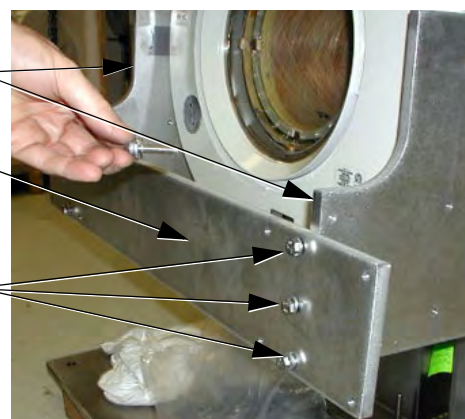
Leaving the washers off the middle bolt provides clearance for the jack pivot.

Note: Before mounting the wheel dollies in the next step, measure the distance from the mounting studs in the floor of the exam room to the wall at the service end of the magnet. If the distance is less than 30 inches (762 mm), install the wheel jacks at the rear of the magnet on the side plates, instead of on the end plate.

Dolly side plate

Dolly end plate

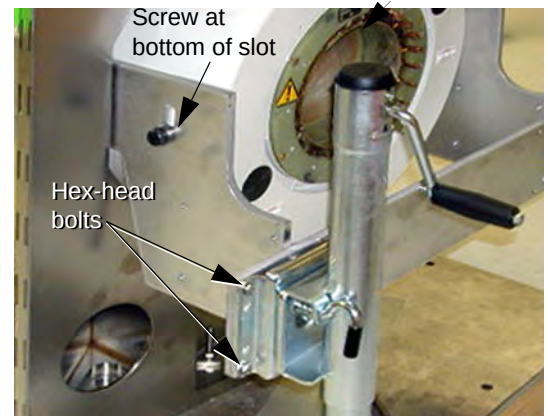
M8 hex-head bolts



Measure any doorways between the exam room and the shipping stand, to verify that they are wide enough for the side-mounted wheel jacks to clear. If the doorways are too narrow, mount the wheel jacks on the end plates and contact Service headquarters, so they can send you the special modified wheel jack.

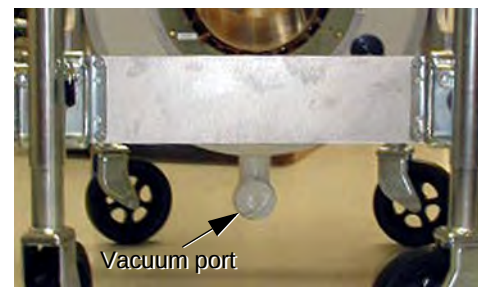
Figure 2-6: Wheel Jack Mounting

4. Attach a wheel jack at each corner of the dolly. Use four M8x25 mm hex-head bolts, with lock washers and flat washers, to mount each wheel jack on the dolly end plates. (See the preceding note.)
5. Turn the crank handles on the two rear jacks to slowly raise the rear of the dolly until the socket-head screws holding the dolly side plates to the magnet are resting at the bottoms of the two slots (see [Figure 2-6](#)). Tighten all four socket-head screws.

**Figure 2-7: Vacuum Port**

6. Raise the magnet by adjusting each wheel jack equally until the magnet is high enough for the vacuum port (see [Figure 2-7](#)), to clear the base of the shipping crate.
7. Remove the base of the shipping crate from under the magnet and move the magnet to the desired location.

Caution: This operation requires two people, to prevent damage to the equipment and site.



Placing the Magnet

Before the magnet can be lowered in place and the dolly removed, the coldhead stand and magnet saddle must be secured to the floor. As part of the site preparation, the customer must anchor studs in the floor for mounting the coldhead stand.

To secure the coldhead stand and magnet saddle:

1. Examine the threads of the coldhead stand mounting studs, and verify that they are not damaged.

The studs are one of three sizes, depending on the site:

- M10
- 3/8-16 UNC
- 3/8-24 UNF

Nuts to fit these stud sizes are in the enclosure kit (2001-0013).

2. Wrap tape around the studs in the floor, to protect the threads.
3. Carefully lower the coldhead stand (2001-1017) onto the studs in the floor. Check to make sure the floor has no bumps that cause the stand to rock.
4. Install the saddle stand (2001-1000) on the coldhead stand (see [Figure 2-8](#)), using two 10 mm hex-head bolts, (1045-0003), two 10 mm lock washers (1047-0036), and two 10 mm flat washers (1047-0035). Leave the bolts hand-tight until the magnet is aligned with the coldhead plate.
5. Remove the tape from the stud threads, and install nuts to secure the stands to the floor.
6. Remove four 3-inch (76 mm) square pads (2001-1040-000) from the enclosure kit. Stack two pieces together and bond the corners with super glue (cyanoacrylate adhesive 1082-0013).

Caution: Use super glue very sparingly. Do not allow any glue to touch your skin.

Caution: Do not use any other adhesives; they can act as a lubricant and allow the pads to slip.

7. Repeat the process with the remaining two 3-inch (76 mm) square pads.
8. Place the 3-inch (76 mm) square pad assemblies on the saddle stand so that they are even with the edge of the stand (see [Figure 2-8](#), above).
9. Remove four 1.5-inch (38 mm) square pads (2001-1040-001) from the enclosure kit.
10. Place one of the 1.5-inch (38 mm) square pads on each of the leveler feet (see [Figure 2-8](#), above).

Figure 2-8: Saddle Stand and Coldhead Stand

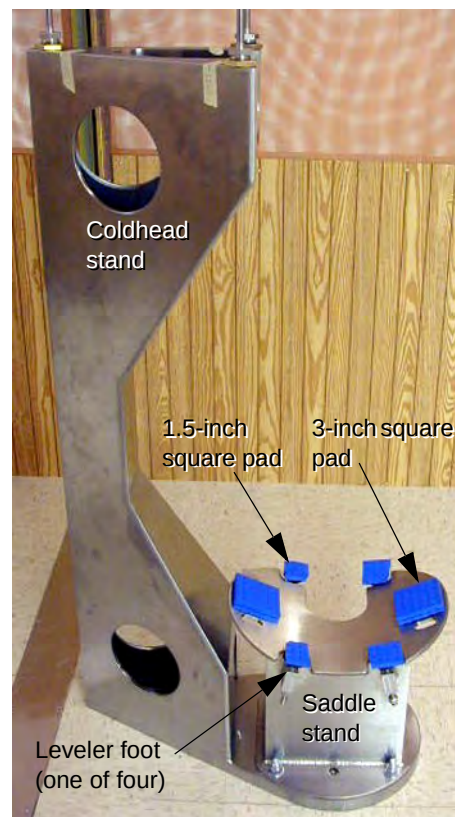


Figure 2-9: Magnet in Place Over Stand

11. Carefully move the magnet into position over the saddle stand (see Figure 2-9). Do not lower the magnet onto the stand yet.

Note: If there is not enough clearance behind the magnet, remove the wheel jacks from the rear end plate and install them on the side plates, or contact Service headquarters to have them send you the special modified wheel jack.



Note: There are two types of vibration mounts. In the next step, check the part number to make sure you are using the correct parts.

12. From the enclosure kit, remove four vibration mounts (1085-0001), four 10 mm hex-head bolts (1045-0004), four 10 mm lock washers (1047-0036), and four 10 mm flat washers (1047-0035).

The vibration mounts come in two sections, male and female. Only the male ends are used; discard the female washer part.

13. Install the four vibration mounts on the magnet saddle. Do not tighten the four hex bolts.

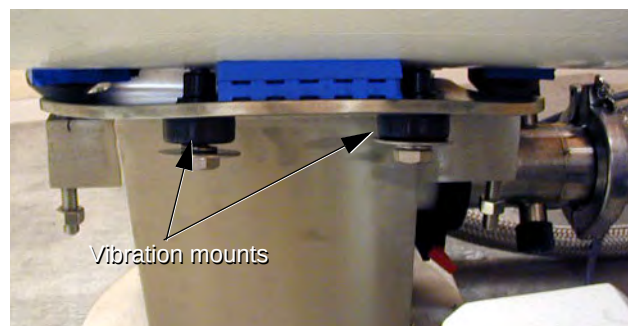


Figure 2-10: Magnet Mounts

Aligning the Coldhead

This procedure covers installing the coldhead plate and aligning the coldhead inside the magnet bellows. Proper coldhead alignment prevents vibration in the magnet, which can cause excessive ghosting during imaging.

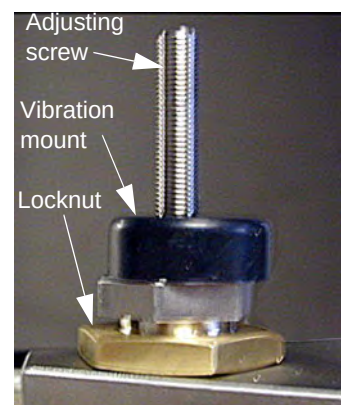
Installing the Coldhead Plate

1. From the installation kit, install three adjusting screws (2001-1021), three locknuts (2001-1022), and the washer parts of three vibration mounts (1085-0007) into the three locations on top of the coldhead stand.
The male part of the vibration mounts are not used; discard them.
2. Apply anti-seize compound to the threads of the adjusting screws.
3. Remove the shipping cover from the top of the magnet bellows.
4. Stuff the opening of the magnet bellows with a clean piece of bubble-wrap or other flexible material.

Caution: Make sure you do not drop anything into the magnet.

5. Remove the O-ring that is installed in the bellows, and replace it with O-ring (1063-0008). Grease the new O-ring sparingly with vacuum grease from the [Vacuum Pump Tool Kit](#)

Figure 2-11: Lower Coldhead Mount



6. Install the coldhead support plate (2001-1018) by centering the three adjustment screws in the three holes of the support plate. Carefully centering the plate will result in more room for adjustments later.
7. Mount the support plate to the coldhead stand using the washer parts of three vibration mounts (1085-0007) (as in [step 1](#)), three flat washers (1047-0034), three lock washers (1047-0033), and three 3/8-24 hex nuts (1048-0013). Hand-tighten the top nuts.

Figure 2-12: Complete Coldhead Mount

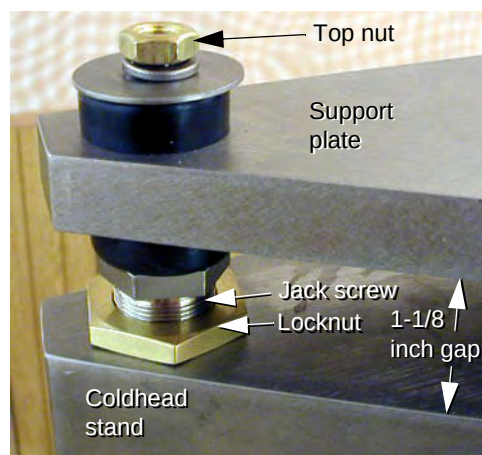


Figure 2-13: Oversize Washer

Note: If the flat washers supplied in the installation kit are large fender washers like the one under the top nut in [Figure 2-12](#), one of the washers may interfere with the coldhead base. Replace the washer on the left-front adjusting screw (shown in [Figure 2-13](#)) with a smaller 10 mm flat washer (1047-0035).

8. Adjust the three jack screws to produce a gap of approximately 1-1/8 inches (28 mm) between the coldhead stand and the support plate at all three locations (see the figure).

You may have to loosen the three top nuts and lift the plate slightly to adjust the three jack screws.

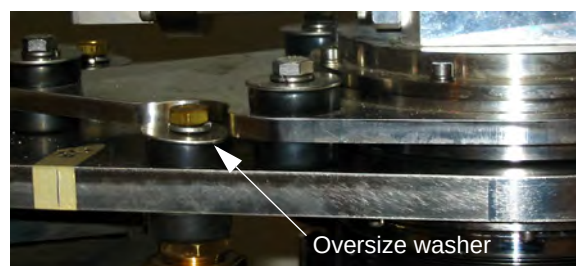
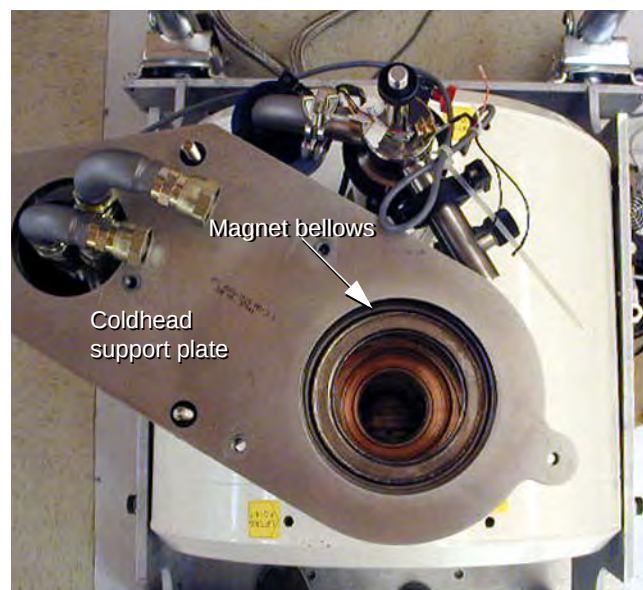


Figure 2-14: Coldhead Support and Magnet Bellows

9. Use the wheel jacks on one side of the magnet to rotate the magnet so the magnet bellows is roughly aligned with the hole in the coldhead support plate.
 - If the magnet bellows is too close to the coldhead stand, raise the two wheel jacks on the left side of the magnet by 1-1/2 turns, then move the dolly toward the coldhead stand. The magnet rotates away from the stand.
 - If the magnet bellows is too far from the coldhead stand, raise the two wheel jacks on the right side of the magnet by 1-1/2 turns, then move the dolly away from the coldhead stand. The magnet rotates toward the stand.

Caution: When rotating the magnet, make sure that the six blue pads stay in place. You may have to raise the magnet slightly, so that the magnet does not drag the pads with it when it moves.

10. When the bellows and the support plate are aligned, slowly lower the magnet onto the saddle.



11. Verify that all six pads are still properly in place, that the two 3-inch (76 mm) pads are still lined up with the edge of the saddle on both sides, and that they are not loose.

Most of the weight of the magnet should rest on the 3-inch (76 mm) square pads at the edge of the saddle. If necessary, raise the stand leveler feet (see [Figure 2-8](#) on page 14) up against the magnet and tighten the locking nuts.

Figure 2-15: Attaching Bellows to Coldhead Support Plate

12. Attach the bellows to the coldhead support plate (see [Figure 2-15](#)), using eight M4x20 mm socket-head screws (1041-0017), eight M4 flat washers (1047-0017), and eight M4 split lock washers (1047-0018).

Install the screws in alternating locations until all eight have been installed, then tighten the screws.

Note: If the screws do not line up with the coldhead support plate, loosen the bolts in either the saddle stand or the coldhead stand, and align the holes.

13. Verify that the three clearance holes in the coldhead plate are centered, and that the bellows is aligned side to side.

Note: Alignment is complete when the following conditions are met:

- The bellows is centered in the support plate
- The vibration mount adjusting screws are centered in the holes of the support plate
- All the weight of the magnet is on the pads of the saddle stand

14. Remove the magnet dolly from the magnet by reversing the procedure [Installing the Magnet Dolly](#) on page 12.

Aligning the Coldhead Plate

1. Remove the stuffing you placed in the magnet bellows earlier.

Caution: Make sure you do not drop anything into the magnet.

Figure 2-16: Coldhead Plate Alignment Tool

2. Insert the coldhead plate alignment tool (see [Figure 2-16](#)) into the bellows.
3. Use feeler gauges to measure the gap between the alignment tool disk and the coldhead mounting plate. Measure around the entire disk.
4. Adjust the three coldhead mounting plate jack screws (shown in [Figure 2-12](#) on page 16) so the gap is uniform around the entire disk.
5. Lock the jack screws by tightening the locking nuts.



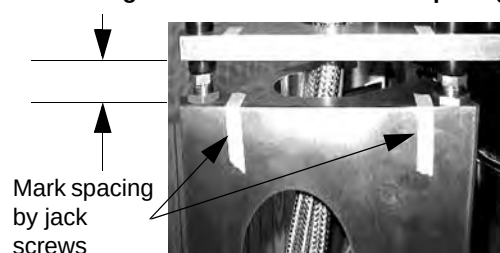
Figure 2-17: Aligning the Coldhead Plate

6. Move the coldhead support plate so that the edge of the disc is flush with the edge of the support plate (see [Figure 2-17](#)).
7. Hand-tighten the three top nuts on the coldhead plate mounts (shown in [Figure 2-12](#) on page 16), to place the isolators under load.
8. Use feeler gauges to measure the gap between the alignment tool disk and the coldhead mounting plate around the entire disk.
9. If the gap is not uniform, repeat steps [step 4](#) through [step 7](#).
10. Remove the coldhead plate alignment tool from the bellows.



11. Mark the distances between the coldhead mounting plate and the top of the coldhead stand on tape near the three jack screws (see [Figure 2-18](#)).
12. Tighten the coldhead stand bolts at the base of the magnet.

Figure 2-18: Plate to Stand Spacing



Installing the Coldhead

Note: All O-rings in the MR system and in the pumping equipment must be lubricated with vacuum grease to prevent leaks.

1. Remove O-ring (1063-0005) from the enclosure kit, apply vacuum grease sparingly to the O-ring, and install the O-ring on the coldhead assembly.
2. Insert the coldhead into the magnet bellows, being extremely careful not to damage the inner sidewalls of the bellows.
3. Mount the coldhead, using four 3/8–24×2 hex-head bolts(1045-0002), four 3/8 lock washers (1047-0036), and four 3/8 flat washers (1047-0034) from the enclosure kit.
4. **CRITICALLY IMPORTANT!** Apply anti-seize compound to the bolts and install them.
5. Tighten the bolts in an alternating pattern.
6. Verify that the coldhead is properly aligned by moving the magnet with both hands, as shown in the figure at right.

The bellows should move considerable amounts (≈ 0.8 inch, or 2 mm) before making contact with the coldhead. The magnet itself should also move. If the magnet feels tight, and you can make contact with the coldhead with little effort, you must repeat [Aligning the Coldhead](#) on page 15.

Figure 2-19: Checking Coldhead Alignment



Caution: It is very important that this alignment is correct. If the coldhead touches the bellows during operation, vibration transmitted to the magnet can cause ghosts in the image.

Pumping out the Magnet

This procedure describes pumping out the magnet in preparation for cooling the magnet. This process ensures that the vacuum space is evacuated to provide thermal insulation for the magnet.

Taking Out the Vacuum Pump Tools

Open the vacuum pump tool kit (see [page 84](#)), and remove its contents.

Removing Shipping Stops

1. Slowly (to carefully release the pre-pumped vacuum) remove one of the four vacuum plugs from the magnet port, using the tool in the [Ramp Tool Kit](#). See [Figure 2-20](#).

Note: Process and replace each plug before removing the next one. Having more than one plug removed can produce undesirable airflow through the vacuum space.

Note: The removal tool may vary depending on the pump kit used.

Figure 2-20: Removing the Vacuum Plugs



2. Loosen, but do not remove the two **socket-head screws** in the magnet port.
3. Screw the long threaded tool into one of the four aluminum disks (1053-0010) from the magnet kit, as shown in the figure at right.
4. Insert the disk into the magnet port and rotate the disk to line up the holes with the two socket-head screws.
5. Tighten the two socket-head screws and remove the installation tool.

Figure 2-21: Installing Disk in Magnet Port



6. A small roll of foil is stored next to the disk inside the magnet. Carefully pull the roll through the port to unroll it.
A hex-key wrench is helpful when unrolling the foil.
7. Cut off a section of silver tape (in the [Vacuum Pump Tool Kit](#)) and make a loop with the sticky side on the outside. Press the loop of tape onto the aluminum disc.
8. Unroll the foil and place it over the aluminum disc.
The loop of tape should hold the foil in place.

Figure 2-22: Foil in Magnet Port



9. Unscrew the stop screw from the vacuum plug (see [Figure 2-20](#)). Thoroughly clean the threads of the vacuum plug and the magnet port. Remove and thoroughly clean the O-ring from the plug. Grease the O-ring sparingly and reinstall it on the vacuum plug. Apply a light coat of anti-seize compound to the threads of the vacuum plug. Be careful not to get anti-seize compound on the O-ring.

Note: Cleaning and lubricating the threads and O-ring may not always be necessary, if they are clean and lubricated. However, performing this step now can prevent much work later.

Figure 2-23: Magnet Vacuum Plug



10. Install the vacuum plug in the magnet, and tighten it.

The plug should thread in by hand until it is almost flush with the face of the magnet. The tool should rotate the plug only about another half-turn. Excess rotation indicates that the O-ring is pinched; remove the plug and inspect the O-ring.

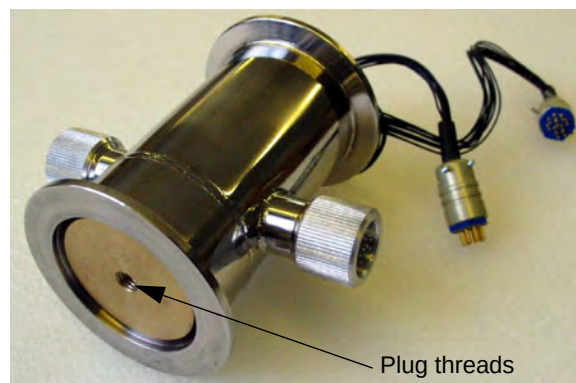
11. Repeat steps 3 to 10 on the other three magnet ports.

Checking the Vacuum Valve

1. There is a vacuum valve installed on the magnet vacuum port (Figure 2-25 shows the port with the valve removed).

- Disconnect the valve from the system.
- Thoroughly inspect and clean the valve assembly.
- Verify that the threads in the plug are clean and are not damaged in any way.
- Apply vacuum grease to the O-rings and proceed with step 2 to reinstall the valve assembly.

Figure 2-24: Vacuum Valve



- Carefully pull the electrical connectors for the shield and the link out of the vacuum port, far enough to verify that they are properly connected.
- Verify that the hydrogen getter bag in the port is intact and will not interfere with pumping out the magnet.

If necessary, reposition the getter bag in the back of the vacuum port.

Note: If the getter bag is located in the vacuum valve, remove it and place it in the vacuum port as shown in the figure at right.

Figure 2-25: Getter Bag in Magnet Vacuum Port

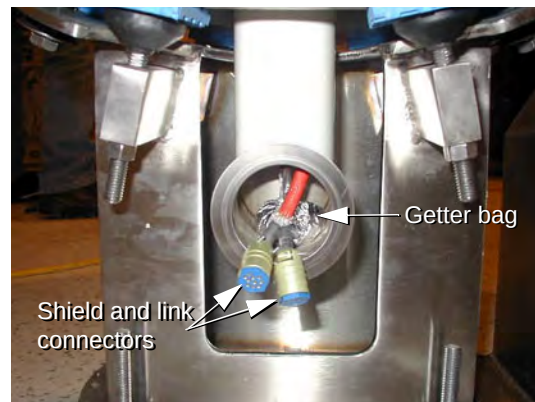


Figure 2-26: Shield and Link Connections at Vacuum Valve

- If needed, re-grease the O-ring of the NW50 sealing ring (1063-0012) sparingly with vacuum grease, and place the ring on the end of the vacuum valve.
- Connect the vacuum valve to the magnet vacuum port with the **clamp** (1063-0013). Position the interface external connectors at horizontal positions.



Assembling the Pumps

Refer to [Figure 2-27](#) and [Figure 2-28](#) below, to identify the connections between the pumping system components.

Figure 2-27: Pump-Out Assembly Connections

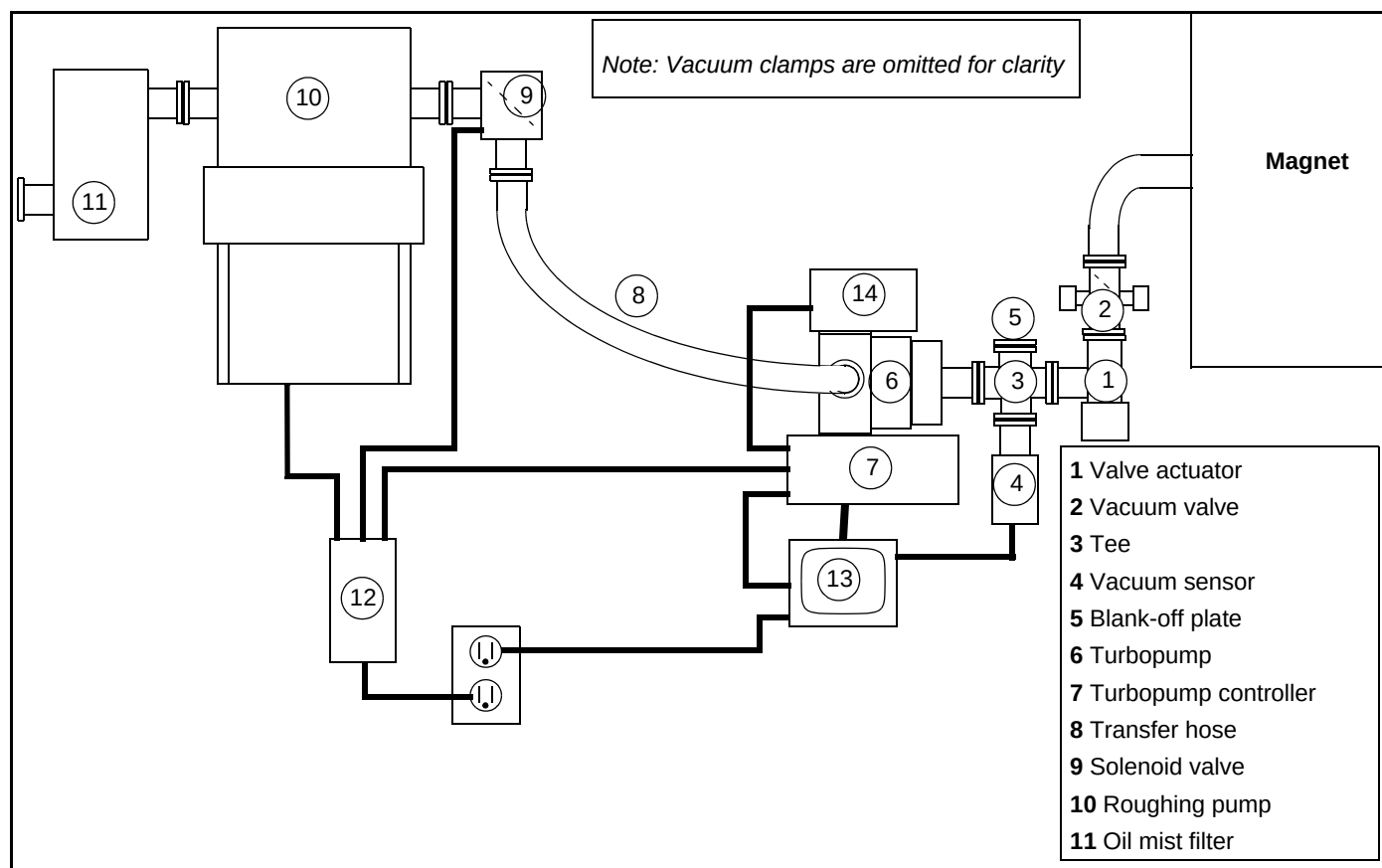
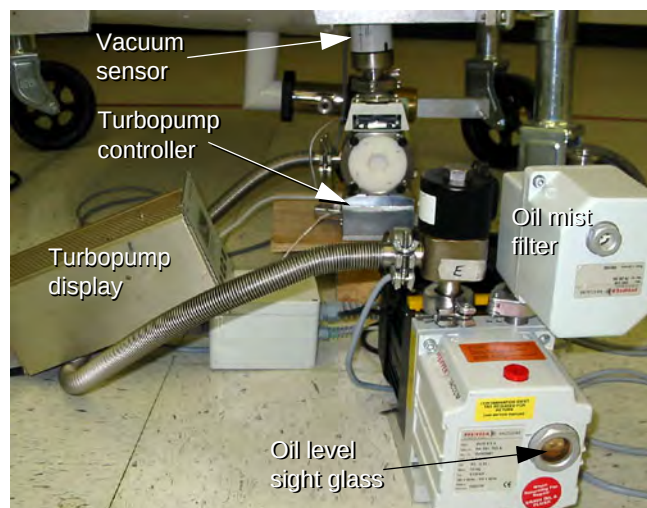
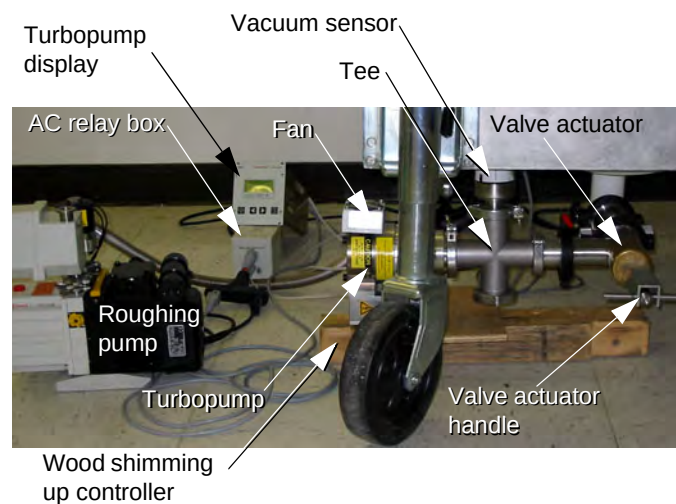


Figure 2-28: Pump-Out Assembly



1. Connect the valve actuator (1) to the vacuum valve (2) with a sealing ring and clamp.
2. Connect the vacuum tee (3) to the valve actuator, using the NW50 to NW40 adapter and sealing rings and clamps.
Rotate the tee so that the two side ports are vertical.
3. Connect the vacuum sensor (4) to the upper side port of the tee with a sealing ring and clamp.
4. Attach a blank-off plate (5) to the lower side port of the tee with a sealing ring and clamp.

5. Connect the inlet port of the turbopump (6) to the remaining port of the tee with a sealing ring and clamp. Optionally, you can use the NW40 corrugated transfer hose between the turbopump and the tee.
6. Rotate the turbopump assembly so that the turbopump controller (7) is parallel with the floor. Shim under the controller so that the weight of the pump is supported by the controller.

The pieces of wood in [Figure 2-28](#) on page 21 are shimming up the controller and turbopump.

7. Connect a transfer hose (8) to the output port of the turbopump with a sealing ring and clamp.
8. Connect the other end of the transfer hose to the solenoid valve (9) inlet port with a sealing ring and clamp.
9. Connect the outlet port of the solenoid valve to the roughing pump (10) inlet port with a sealing ring and clamp.
10. Connect the oil mist filter (11) to the roughing pump outlet port with a sealing ring and clamp.
11. Make the electrical connections between the components as shown in [Figure 2-27](#) on page 21.
 - a. Connect the power cords of the roughing pump, and solenoid valve to the AC power relay box.
 - b. Connect the RJ11 connector cord coming from the AC power relay box to the second RJ11 connector from the right on the turbopump controller.
 - c. Connect the power cords of the AC power relay box and turbopump display to a 110 VAC building power outlet.
 - d. Connect the signal cable of the vacuum sensor to the turbopump display (13).
 - e. Connect the RS485 data cable between the turbopump display and the turbopump controller.
 - f. Connect the combined control and power cable from the turbopump display to the turbopump controller.
 - g. Connect the RJ11 connector power cord of the cooling fan (14) to the rightmost RJ11 connector on the turbopump controller.

Starting the Pumps

1. Examine the roughing pump to verify the following conditions:
 - The sight glass shows an adequate oil level.
 - The yellow gas ballast switch is in the OFF position (0 lined up with 0)
 - The roughing pump power switch is in the ON position. This switch can be difficult to locate, but is typically on the box where the power cable enters the pump.

2. Verify that the mist filter has little or no oil in it.

If there is significant oil in the mist filter, refer to the *Filter Service Manual* for instructions on how to drain it.

3. Verify that the pressure relief valve on the turbopump is closed.
4. Screw the valve actuator handle (see [Figure 2-28](#) on page 21) into the vacuum valve and open the valve by slowly pulling the handle towards you. Lock the valve open.
5. Turn on the switch at the rear of the turbo controller display unit.

The solenoid valve snaps open, and the roughing pump starts up. The turbopump should come up to speed in about 2 minutes, and the vacuum should quickly drop into the 10^{-2} mbar range. If the pumps do not start up, switch places of the two RJ11 cables on the turbopump controller.

Note: Use the arrow keys on the turbo controller display to scroll to 340 to display the vacuum gauge reading. You can also check (scroll to 309) that the turbo pump is spinning at 1500 rpm.

6. Run the pumps approximately 30 minutes to partially pump down the magnet. This prevents the vacuum valve from pulling open during the next operation.

Figure 2-29: Turbopump Relief Valve



Figure 2-30: T-Handle in Valve Actuator

Close the vacuum valve by rotating the T-handle to unlock it, and pushing the handle in.

The vacuum should reach the 10^{-6} mbar range within two hours. If the vacuum does not drop, there is probably a leak in the plumbing. Check all of the connections and, if necessary, clean and re-grease the O-rings.

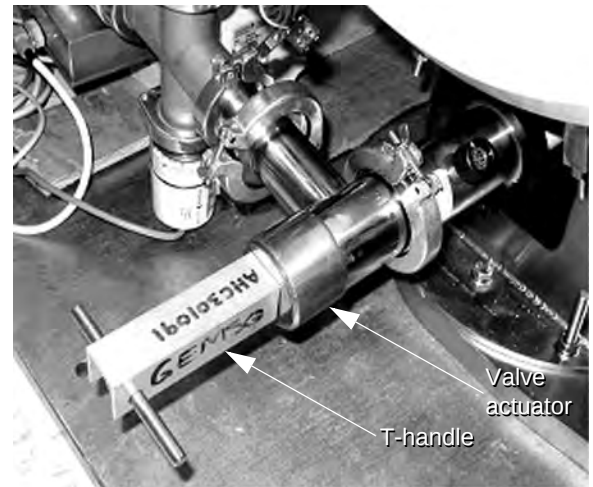
7. When a good vacuum is established and the integrity of the plumbing is verified, shut down the pumps.
8. Open the relief valve on the turbopump until you hear a slight hiss.
9. When the pump assembly is at atmosphere, close the turbopump relief valve.

The assembly is at atmosphere when the turbopump relief valve stops hissing.

10. Open the valve on the magnet by slowly pulling the valve actuator handle toward you.
11. Set the handle horizontally to hold the valve open (as shown in [Figure 2-30](#)).
12. Turn on the roughing pump and the turbopump.
13. Run the pumps for 24 to 48 hours, until the vacuum reaches approximately 10^{-5} mbar.

Note: If the pumps don't produce adequate vacuum level, clean and re-grease the magnet vacuum plug O-rings (see [Figure 2-23](#) on page 19).

14. Begin the next section, [Equipment Installation](#) while the magnet pumps down.



Equipment Installation

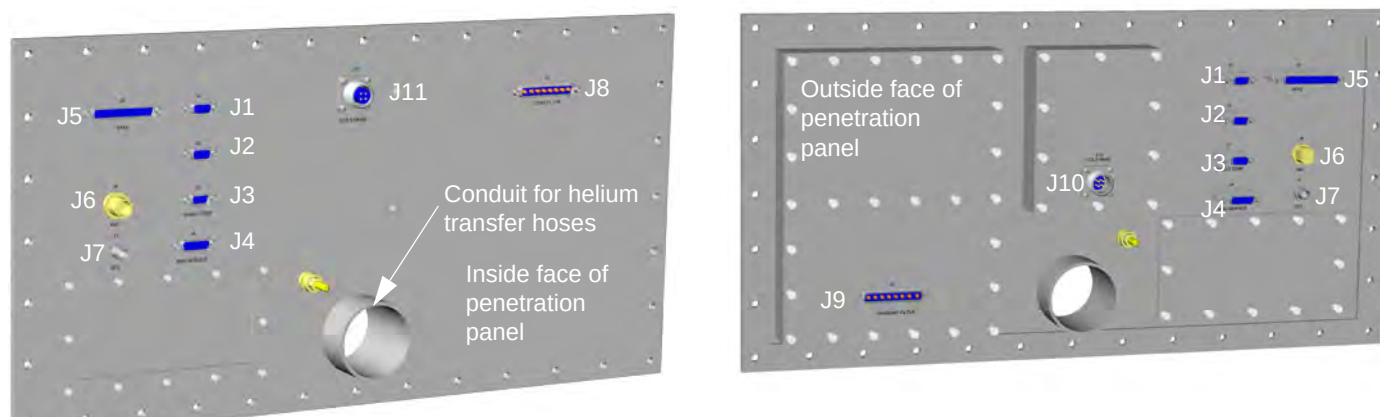
This section covers installation of the following equipment:

- Installing the Compressor on page 24
- Installing the RF Coil Interface Plate (the “G10 Plate”) on page 25
- Installing the Pressure Sensor Assembly on page 26
- Installing the Quench Vent and Hose on page 27
- Connecting the Cable Harness on page 27
- Connecting the Cable Harness on page 27
- Installing the Compressor on page 24

Installing the Compressor

1. Insert the penetration panel from the equipment room side of the exam room wall.

Figure 2-31: Penetration Panel



2. Install 36 1/4-inch lock washers (1047-0037), and 36 1/4-inch flat washers (1047-0028) on 36 1/4-20x2-1/2-inch screws (1042-0006)
3. From inside the exam room, insert the **screws** through the wall and thread them into the holes around the perimeter of the penetration panel.
4. Remove the compressor shipping cover and lift the compressor off its pallet.

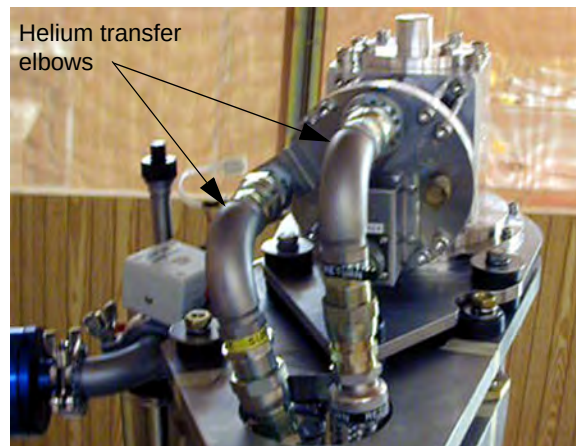
WARNING: The compressor is heavy. At least two people are required to lift it.

5. Roll the compressor to its operating location.
6. Connect the 220 VAC power cord to the compressor.
7. Install the two 90-degree helium transfer elbows (1099-0033-001 and 1099-0033-000) on the coldhead, using the special orange-handled wrench in the pump kit.

The supply elbow is labeled in yellow. The return elbow is labeled in green.

Caution: When securing helium transfer fittings, use a second wrench to hold the male fittings to prevent their loosening.

Figure 2-32: Coldhead with Transfer Hoses



8. Run the helium transfer hoses from the compressor to the coldhead. Feed the hoses through the conduit in the penetration panel (see [Figure 2-31](#)) and up through the magnet stand. Stuff the penetration panel conduit with bronze wool (1063-0007) around the hoses. Use gloves when handling the helium transfer hoses and the bronze wool to avoid metal splinters.

Caution: Do not bend the hoses too tightly. The minimum bend radius is about one foot.

9. Secure the transfer hoses to the elbows.
10. Secure the other ends of the transfer hoses to the compressor, using two wrenches.
Use one wrench to turn the fitting nut at the end of the transfer hose, and another wrench to hold the fitting on the compressor.

Installing the RF Coil Interface Plate (the “G10 Plate”)

1. Record the gradient coil serial number from the nameplate on the rear of the magnet. Write the number on a piece of tape applied to the magnet above the RF coil interface plate location.

This number is required for [Chapter 3, System Configuration and Test](#), and for PM procedures. Installing the RF coil interface plate covers the nameplate.

2. Use a heavy-duty abrasive pad (Scotchbrite® or equivalent) to remove any coating from the gradient coil lugs on the RF coil interface plate, and examine the holes in the lugs for epoxy. If there is any epoxy in the holes, remove it, or the cable bolts will not fit.

Caution: Varnish or other coating on the gradient coil lugs can cause gradient amplifier overload.

3. Mount the RF coil interface plate and the bulkhead connector bracket (2001-1023) on the magnet, using three M6x25 socket-head screws (1041-0013), three M6x12 socket-head screws (1041-0015), six M6 lock washers (1047-0015), and six M6 flat washers (1047-0006). (See [Figure 2-33](#) for screw locations) Use the bulkhead connector hole closest to the bend. Do not tighten the screws; the plate should be loose.
4. Insert one of the coils in the bore of the magnet, and mate the coil to the plate. Verify that the coil lines up properly and latches to the plate.
5. Tighten the screws to secure the RF coil interface plate to the magnet.
6. Unlatch the coil and remove it from the magnet, then install the coil and verify that the alignment is proper, and that the coil latches properly.

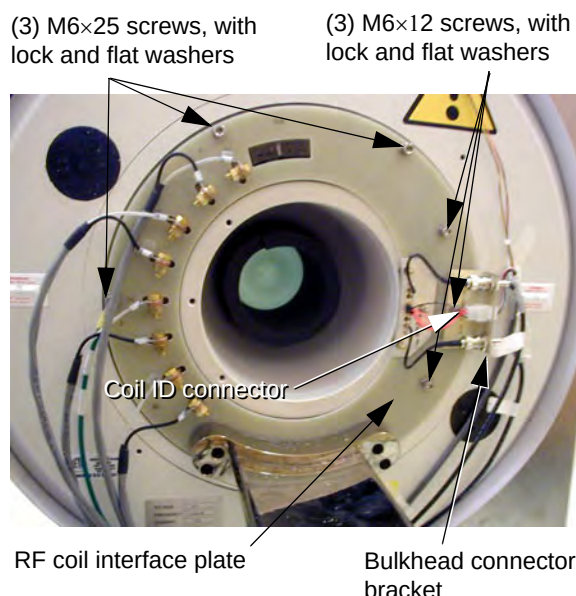
You should be able to install and remove the coil easily, and it should not be tight.

7. Install the other coil and verify its fit in the same manner. You may have to loosen the RF coil interface plate screws and readjust the plate to achieve a proper fit with both coils.

Note: If the proper fit cannot be achieved, call Service support (see [Trademarks](#) on page 9). Never disassemble the RF coil interface plate without direction from GE engineering. The brass dagger pins are mounted using LocTite and will break if you try to unscrew them.

8. Install the nylon screw to secure the bulkhead bracket. Use the hole furthest away from the bend of the bracket.

Figure 2-33: RF Coil Interface Plate



9. Connect the I and Q cables from J3 and J4 (respectively) on the RF front end to the RF Interface plate, positioning the leads as shown in the figure at right.
10. Connect the Coil ID/laser switch cable.
11. Apply a cable tie anchor to the magnet and secure the three cables, using a cable tie to provide strain relief.
12. Connect the gradient coil cables from J8 on the penetration panel to the coil lugs, as shown at right, and secure them to the magnet using a cable tie anchor and cable tie.

Figure 2-34: I, Q, and Coil ID Cables

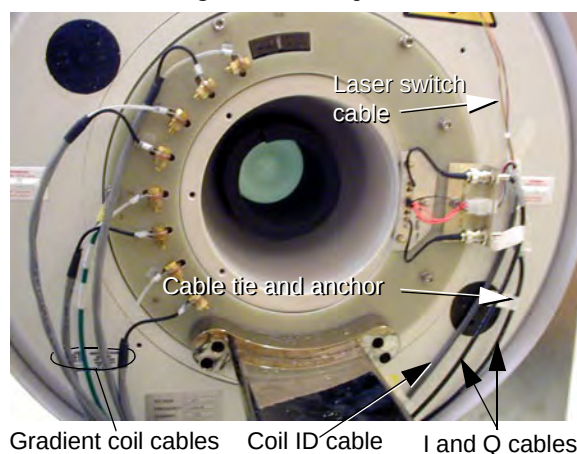


Figure 2-35: Pressure Sensor

Installing the Pressure Sensor Assembly

1. Install the pressure sensor assembly (2001-0004) on the magnet upper vacuum tee, using a sealing ring and clamp (1063-0003). The pressure sensor should be mounted sticking into the magnet stand.

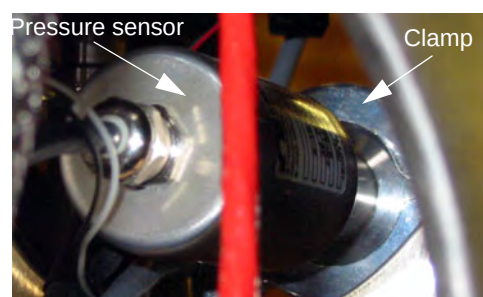


Figure 2-36: Teflon Tape on Pressure Sensor

2. Verify that there is teflon tape on the sensor pipe threads. If tape is not visible, remove the sensor, clean the threads, apply teflon tape to the threads, and install the sensor.
3. Verify that the sensor is parallel to the port on the magnet. Verify that it is properly seated and not skewed, to prevent helium leaks.

If the sensor is not mounted properly, remove it and install it properly.

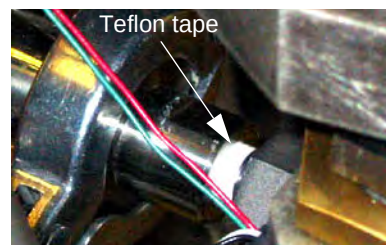


Figure 2-37: Blank-Off Plate

4. Install a NW25 blank-off plate (1062-0009) on the magnet upper vacuum tee on the opposite side from the pressure sensor, using a sealing ring and clamp.

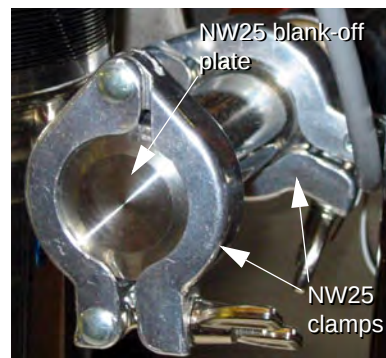


Figure 2-38: Quench Valve



Installing the Quench Vent and Hose

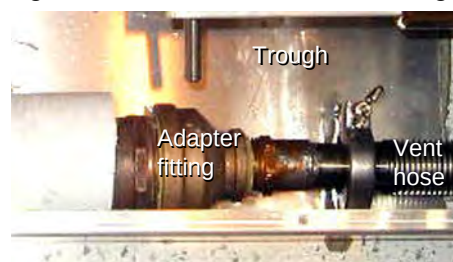
1. Place a sealing ring on the 90-degree elbow (1062-0010), and attach the elbow to the service turret using an NW25 clamp.
2. Place a sealing ring on the other end of the 90-degree elbow, and use a clamp to attach the quench vent to the elbow with the arrow on the quench vent pointing down.

Caution: Make sure the arrow on the quench vent points down, as shown in the figure at right.

Position the vent so that it is not touching the coldhead stand. If necessary, loosen the service turret clamp and rotate the turret so that the vent is parallel to, but not touching the edge of the coldhead stand.

3. Attach the flex hose (1067-0005) to the bottom of the vent, using a sealing ring and clamp.
4. Connect the other end of the flex hose to the brass adapter fitting (1062-0011) using a sealing ring and clamp. Position the hose in the cable trough. The trough and brass adapter fitting are installed by local contractors.

Figure 2-39: Vent Hose in Cable Trough



Connecting the Cable Harness

1. Continue to install the system interconnect cables to the locations specified on the label of each cable.
2. Use M6 brass hex-head screws (1045-0005), M6 brass flat washers (1047-0038), M6 brass lock washers (1047-0015), and M6 brass hex nuts (1048-0017) to connect the cable harness from the penetration panel to the rear of the magnet, matching the gradient and polarity on the cable label to the locations marked on the RF coil interface plate.

Removing the Pump Assembly

1. Check the magnet vacuum shown on the turbopump display.
2. When the magnet vacuum is acceptable (approximately 10^{-5} mbar), close the vacuum valve so that the plug is firmly seated.
Verify that the plug is seated by monitoring the turbopump controller display for any sudden drop in pressure.
3. Release the valve from the magnet by turning the T-handle on the vacuum valve counterclockwise.

As you rotate the valve actuator handle, observe the handle shaft where it enters the actuator. You should see the shaft backing out. This is easier if you put a mark on the shaft. Keep rotating the valve actuator handle until it no longer backs out. You may be able to feel the shaft bouncing on the threads as you rotate it counterclockwise.

To verify that the plug is firmly seated and released from the valve actuator handle, pull out the valve actuator handle while observing the turbopump controller display.

- If the pressure momentarily increases slightly, then drops back to the value displayed after seating the plug, the plug is properly seated; continue with the next step.
- If the pressure rises suddenly, the plug is not released from the valve actuator handle, and the valve is open. Repeat this step until you are sure the plug is seated.

Caution: Do not continue with the removal until you are sure the plug is seated.

4. When you are sure the plug is firmly seated and released from the T-handle, shut off the pump assembly with the switch at the rear of the turbopump display.
5. Slowly open the pressure relief valve at the rear of the turbopump, and allow the assembly to come to atmosphere.
6. Allow the turbopump to spin down. This can take several minutes.
7. Remove the pump assembly from the magnet.
8. Install the plastic cap that was shipped on the magnet vacuum port over the vacuum valve.
9. Return the vacuum pump tools to their container.

Powering up the System

Before cooling the magnet, you must verify that the coldhead and the compressor are operational, and that the helium-fill monitoring software functions correctly.

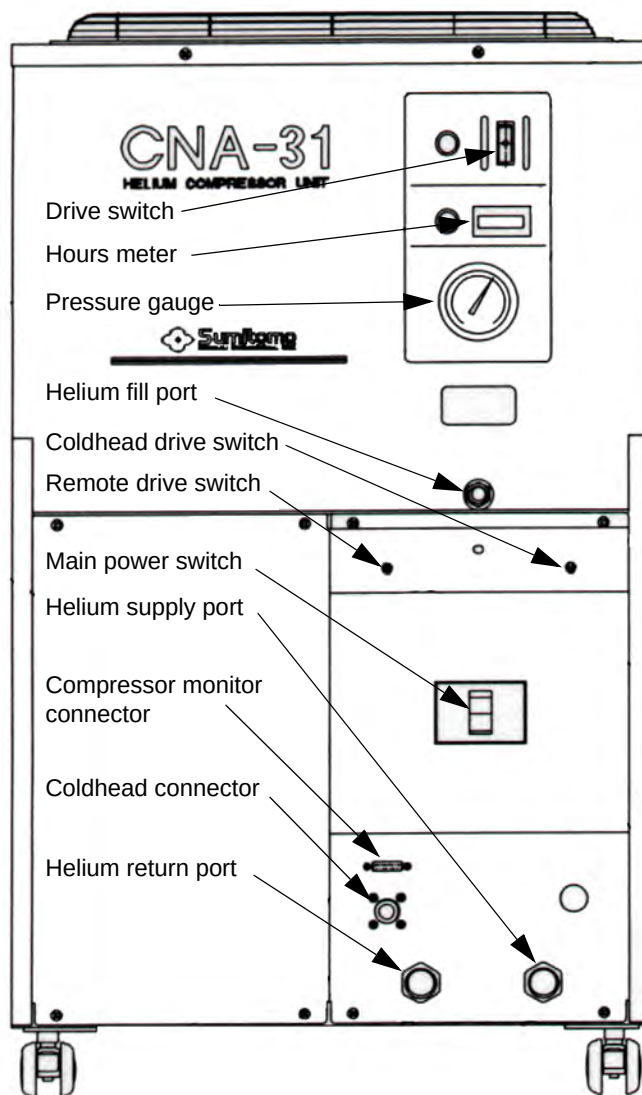
Verifying Logon and Helium Charge

1. Turn on the two **circuit breakers** at the bottom rear of the electronics cabinet.
2. Verify that the system powers up correctly and allows you to log into the **MR Scanner** program. (Refer to the *Operator's Guide* for information on logging on to the ORTHONE Scanner.)
3. Verify that the static pressure on the gauge at the back of the compressor is between 1.6 and 1.65 MPa. (See [Figure 2-40](#) on page 29 to locate features on the back of the compressor.)
 - If the static pressure is less than 1.6 MPa, perform [Charging the Compressor](#) on page 30.
 - If the static pressure is at least 1.6 MPa, skip to [Starting the Compressor](#) on page 31.

Figure 2-40: Rear of Compressor

Note: The *Main Power* switch controls all power to the compressor and coldhead. The *Drive* switch controls power to the compressor and coldhead motors when the Main switch is ON. The *Cold Head Drive* switch controls power to the coldhead motor when the main switch is ON, and the compressor is not running.

Caution: *Do not switch the Cold Head Drive switch to ON. This switch is used only when the cold head must be cycled with the compressor off. With the switch ON, the coldhead can be damaged if the compressor shuts down because of an over temperature or other fault. To operate the compressor and coldhead together, use the Drive switch, not the Cold Head Drive switch. You may want to put a piece of tape over the Cold Head Drive switch to prevent anyone from accidentally activating it.*



Charging the Compressor

Charging the compressor with helium gas is not normally necessary. Only if the static helium pressure reading is below 1.6 MPa should you add helium to the compressor. (See [Figure 2-40](#) on page 29 to locate the pressure gauge.)

Caution: The regulator, the charging line, and the gas charging tool must be purged of air before connecting them to the compressor. If there is air in the fill assembly when it is connected to the compressor, the air will mix with and contaminate the helium. Air in the helium can freeze in the coldhead, causing it to malfunction.

Purging the Gas Fill Assembly

This procedure ensures that there is no air in the helium fill assembly before it attaches to the compressor. The procedure requires a bottle of grade-5 helium gas and the following equipment from the [Transfer Tool Kit](#) (see page 83):

Figure 2-41: Gas Charging Tool

- The helium gas transfer hose
- The gas charging tool shown at right
- A male fitting that mates to the female fitting on the gas charging tool
- A gas regulator that fits the helium gas bottle
- Wrenches to fit the various gas fittings



WARNING: Pressurized gas bottles can be dangerous if they fall over. Always make sure that gas bottles are secured to a wall or a stable cart, or place the bottles horizontally on the floor.

To purge the fill assembly:

1. Connect the regulator to the helium gas bottle.

Caution: Do not open the gas bottle valve yet.

2. Turn the regulator handle in several turns.

This ensures that when you open the gas bottle valve, helium will flow through the regulator.

3. Connect the helium gas transfer hose to the regulator.
4. Connect the gas charging tool to the transfer hose.
5. Thread the male gas fitting into the fitting of the charging tool. After you feel the fitting sealing surfaces meet, turn the fittings together two more turns by hand.
6. Open the charging tool valve.

This ensures that when helium gas enters the fill line, the gas flows continuously out to atmosphere, purging air from the assembly.
7. Open the helium gas bottle valve slightly.

Gas flows out the male fitting attached to the gas charging tool.
8. Allow the gas to flow for one minute.
9. Unscrew the male fitting from the gas charging tool.

The seal of the female fitting closes, stopping gas flow, with pressure in the line.
10. Adjust the regulator to 250 to 300 PSI.
11. Close the charging tool valve.

Charging the Compressor With Helium

To recharge the compressor:

1. Remove the protective cap from the gas charge coupling on the rear of the compressor.
2. Connect the gas charging tool to the charge coupling on the rear of the compressor.

3. Slowly open the charging tool valve, while monitoring the pressure gauge on the back of the compressor. When the indicated pressure reaches 1.60 to 1.65 MPa, close the valve.
4. Disconnect the charging line and gas charging **tool** from the compressor.
5. Install the protective cap on the compressor gas charge coupling.

Starting the Compressor

1. Plug the compressor power cord into the site power outlet and turn the compressor on.

Note: There are three power switches on the back of the compressor. The *Main Power* switch controls all power to the compressor and the coldhead. The *Drive* switch controls power to the compressor and coldhead motors when the Main switch is ON. The *Cold Head Drive* switch controls power to the coldhead motor when the main switch is ON, and the compressor is not running. For normal operation, switch the Main Power and Drive switches to ON, and the Cold Head Drive switch to OFF.

Caution: **Do not switch the Cold Head Drive switch to ON. This switch is used only when the cold head must be cycled with the compressor off. With the switch ON, the coldhead can be damaged if the compressor shuts down because of an over temperature or other fault. To operate the compressor and coldhead together, use the Drive switch, not the Cold Head Drive switch. You may want to put a piece of tape over the Cold Head Drive switch to prevent anyone from accidentally activating it.**

Note: If the compressor does not turn on, incorrect input power phasing is the likely cause. Switching any two of the three incoming phase leads at the compressor power plug should solve this problem.

2. Place your hand on the coldhead and listen for the characteristic chirping noise to verify that the coldhead is operating.
3. With the compressor running, check the pressure gauge for a reading between 1.9 and 2.1 MPa. Do not allow the compressor to run for more than a few minutes.

If the pressure is not in the range of 1.9 to 2.1 MPa, refer to the compressor service documentation for information on troubleshooting the compressor.

4. Shut down the compressor and proceed to [Cooling Down the Magnet](#) on page 31.

Cooling Down the Magnet

The magnet must be very cold for the coils to superconduct. The following processes chill the magnet by steps:

- The first step chills the magnet with liquid nitrogen.
- The nitrogen is then flushed out with helium gas, because any nitrogen remaining in the magnet would freeze in the liquid helium that is the final coolant.
- When all of the nitrogen is removed, the magnet is filled with liquid helium.
- The coldhead and compressor are started, to keep the magnet chilled to 4° K.

Taking Out the Transfer Tools

Open the transfer tool kit (see [page 84](#)), and remove its contents.

Testing the MMU Sensors and Quench Button

This procedure tests the following magnet monitoring unit (MMU) components:

- Emergency quench button
- Helium level sensor
- Link temperature sensor
- Shield temperature sensor
- Battery
- Heater circuit

Hardware Required

This procedure requires the following equipment:

- MMU service test fixture P/N 2001-8000
- DMM

Procedures

Test the Quench Button, Battery, and Heater Circuit

1. Set the toggle switch on the MMU service test fixture to the Q\LHe position.
2. Disconnect the magnet services cable (P/N 2005-3013-002) from the service turret of the magnet and connect the cable to the MMU test fixture.

Figure 2-42: MMU Service Test Fixture



Figure 2-43: Emergency Quench Button



3. Test the emergency quench button:
 - a. Connect a DMM set to measure DC volts to the banana jacks on the MMU test fixture.
 - b. Verify that the DMM reads 0 VDC.
 - c. Push the emergency quench button and record the DMM voltage reading. The reading should be 18 to 24 VDC.
 - d. Push the quench button again to turn off the circuit.
 - e. Turn off the MMU, push the quench button again, and record the voltage reading. The DMM should read 16 to 22 VDC.
 - f. Push the quench button again to turn off the quench circuit, and verify that the DMM reads 0 VDC.

Caution: Make sure the quench button is in the outer position (flush with the edges that surround it) to make sure you don't quench the magnet when you reconnect the magnet services cable to the magnet.

- g. Turn the MMU back on.

- h. Log in to the MR application and enter Service mode.

Figure 2-44: Entering Service Mode

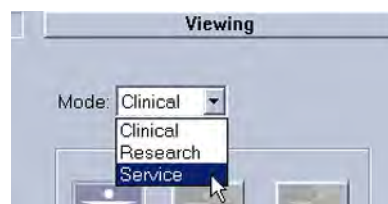


Figure 2-45: Freeing the CAS Bus



- i. Click **Service Tools**, then click **Free CAS**.
The system frees the CAS bus for use by CasWin.
- j. Click **Service Tools**, then click **CasWin**.
The CasWin window opens.
- k. On the CasWin window, click the **MMU1 tab**.
The CasWin window changes to show the MMU sensor fields and other information.

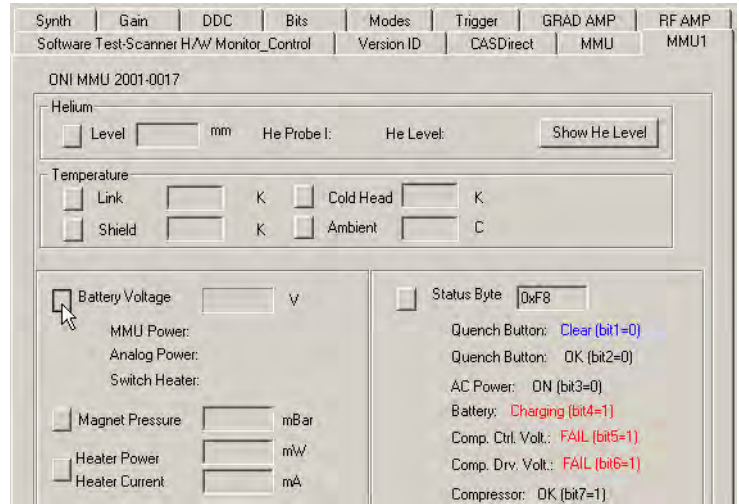
Figure 2-46: Resetting the Quench Bit

1. On the CasWin MMU1 window, click the Reset Quench/Fault Bits button.
CasWin resets the quench bit to the normal state.



Figure 2-47: Reading Battery Voltage in CasWin MMU1 window

4. Check the MMU battery voltage and heater circuit:
 - a. In the CasWin MMU1 window, click the **Battery Voltage** button, and verify that the value is acceptable.
The minimum acceptable voltage is 18 V.
 - b. Click the **Heater Power/Heater Current** button.
 - c. Verify that the **Heater Power** and **Heater Current** values are acceptable. For a new coldhead where the system has had time to stabilize, the heater current is typically 60-80 mA and the heater power 400-500 mW.



Test the Helium Level, Link Temperature, and Shield Temperature Sensors

1. Click the **Calibrate** button and change the following values as needed. Click the **Write** button associated with each set of values after they have been changed.

Table 1: MMU Test Values to Modify

Parameter	Slope	Scale	Offset
Helium Level	2	10	0
Link	3392	10000	914
Shield	3392	10000	914

2. Click **Done** to close the Calibrate window. In the CasWin MMU1 window, click the **Helium Level** button, and read the helium level.
3. Verify that the reading is about 600 mm, and that the **Probe 1** and **He Level** show **PASS**.
4. Disconnect the **MMU test fixture**
5. Connect the **Magnet Services cable** to the magnet service turret connector.
6. Set the **toggle switch** on the MMU test fixture to **TEMPS**.
7. Disconnect the **Shield Temp cable** (P/N 2005-3014-002) from the vacuum port of the magnet and plug it into the **MMU test fixture**.

Figure 2-48: Testing the Helium Level Sensor

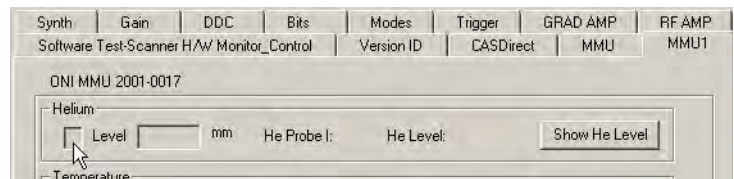
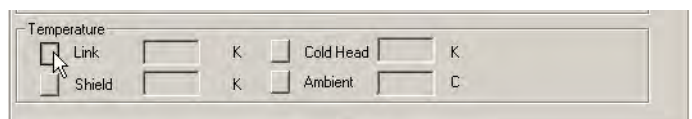


Figure 2-49: Testing the Link Temperature Sensor

8. In the CasWin MMU1 window, click the **LINK** button, and read the magnet link temperature.

The value should be about 10 K.



9. In the CasWin MMU1 window, click the SHIELD button, and read the magnet shield temperature.

The value should be about 10 K.

10. Click the Calibrate button and change the following values as needed. Click the Write button associated with each set of values after they have been changed.

Table 2: 1000291-15 < Magnet serial number < 1000291-51

Parameter	Slope	Scale	Offset
Helium Level	33	100	360
Link	3392	10000	64622
Shield	3392	10000	64622

Note: If the MMU1 is revision E or earlier, the Offset for Link and Shield should be 914 in Table 2.

Table 3: Magnet serial number > 1000291-51

Parameter	Slope	Scale	Offset
Helium Level	33	100	360
Link	1867	10000	31
Shield	1867	10000	31

11. Disconnect the MMU test fixture and connect the Shield Temp cable to the magnet vacuum port.
12. Check the shield and link readings in CasWin to verify that the cable is properly connected.
- The shield and link temperature fields should read about 290 K.
13. Update the helium level reading in CasWin, and verify that the Probe 1 OK radio button highlights.

Characterizing the Magnet

Note: The break-out box makes this procedure much easier to execute.

In this procedure, you use a multimeter to measure the resistance values of the various magnet temperature-sensing circuits, before and during cooling of the magnet.

The difference of two resistance measurements is converted to a temperature, using the tables in Reference Tables on page 78. (The conversion tables give typical conversion values, but systems may differ slightly.)

Measurements taken during this procedure must be entered in the Magnet Characterization Data Sheet.

The notation R_{AB} used below indicates the resistance between pin A and pin B. The pin labels are shown in Figure 2-52.

Figure 2-50: Testing the Shield Temperature Sensor

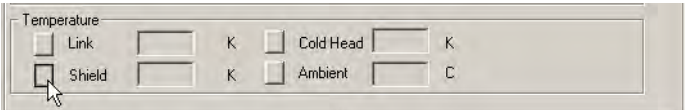
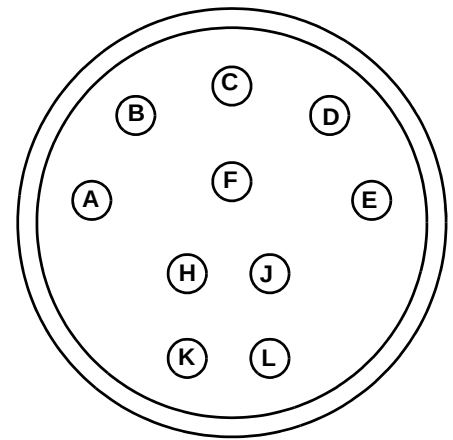


Figure 2-51: Magnet Characterization Data Sheet

Location	Magnet Pump-out Port					
	Coldhead Stand Side			Link		
Sensor	Link		Shield		Link	
Pins	A-D	A-B	H-K	H-J	A-D	A-B
When	Magnet warm / Before liquid nitrogen fill					

Figure 2-52: Magnet Monitoring Pinouts



1. Read the magnet link temperature on the MMU1 screen in CasWin.

If the magnet is at room temperature:

- a. Measure R_{AD} and R_{AB} at the magnet vacuum port connector.
 - b. Subtract R_{AB} from R_{AD} .
 - c. Convert the value to temperature, using Table A-6, [CLTS Resistance-to-Temperature Conversions](#), on page 81.
 2. Record the magnet link values in the *Magnet Characterization Data Sheet*.
 3. Read the magnet shield temperature on the MMU1 screen in CasWin.
 - If the magnet is at room temperature:
 - a. Measure R_{HK} and R_{HJ} at the magnet vacuum port connector.
 - b. Subtract R_{HJ} from R_{HK} .
 - c. Convert the value to temperature, using the CLTS conversion table.
 4. Record the magnet shield values in the *Magnet Characterization Data Sheet*.
 5. Measure the first stage coldhead temperature:
 - a. Measure R_{HK} and R_{HJ} at the coldhead mounting plate port connector.
 - b. Subtract R_{HJ} from R_{HK} .
 - c. Convert the value to temperature, using the CLTS conversion table.
 6. Record the first stage coldhead resistance and equivalent temperature and the temperature read on the MMU1 page in CasWin in the *Magnet Characterization Data Sheet*.
 7. Measure the Allen-Bradley switch temperature:
 - a. Measure R_{HJ} at the service turret connector.
 - b. Convert the value to temperature, using Table A-5, [Allen-Bradley Resistance-to-Temperature Conversions](#), on page 80.
 8. Record the Allen-Bradley switch resistance and equivalent temperature in the *Magnet Characterization Data Sheet*.
- Note:** The Allen-Bradley switch cannot measure temperatures above 100 K and will output values corresponding to 100 K for all higher temperatures.

Filling With Liquid Nitrogen

When operating, the magnet is chilled by liquid helium (LHe), which is expensive. To reduce costs, the magnet is chilled with cheap liquid nitrogen before filling with LHe. The space around the magnet must be pumped down to adequate vacuum (see [Pumping out the Magnet](#) on page 18) before chilling.

WARNING: *Asphyxiation hazard! The nitrogen purge/precool procedures release nitrogen gas, which will displace oxygen in the air and potentially cause asphyxiation. Ensure the following:*

- The magnet room vent system is installed and functioning in conformance with site requirements.
- The magnet room door is propped open.
- A portable floor exhaust fan (>850 CFM or >24 m³/min) is installed and functioning.
- An oxygen monitor is installed and functioning in the magnet room in conformance with site requirements, or that service personnel use properly functioning oxygen monitors in the scan room.

WARNING: *Potential cold burn hazard! You may be exposed to cryogens or contact cold objects during service procedures or a magnet quench. When performing any service procedure that uses cryogens, wear protective clothing:*

- Long sleeve shirt
- Long pants
- Protective apron or jacket
- Non-absorbent gloves
- Glasses or goggles and a face shield

Figure 2-53: Nitrogen Fill Setup

1. Remove the tube nut, the brass washer, and the O-ring from the magnet fill port on the service turret, and place them over the nitrogen fill tube from the transfer kit.
2. Thread the tube into the fill port one to two full turns; it should be tight. Snug down the tube nut.
3. Attach the braided stainless flex hose to the NW-25 flange on the nitrogen fill tube, using a clamp and sealing ring.

Note: The rubber hose shown in [Figure 2-53](#), [Figure 2-55](#), and [Figure 2-57](#) has been replaced with a braided stainless flex hose. Use the braided flex hose.

4. Attach the other end of the braided stainless flex hose to the liquid port of the nitrogen dewar. (See [Figure 2-55](#).)
5. Disconnect both ends of the flex hose that is connected to the quench vent.
6. Remove the quench vent from the service turret and set it aside.
7. Attach the flex hose removed in [step 5](#) to the port where the quench vent was connected.

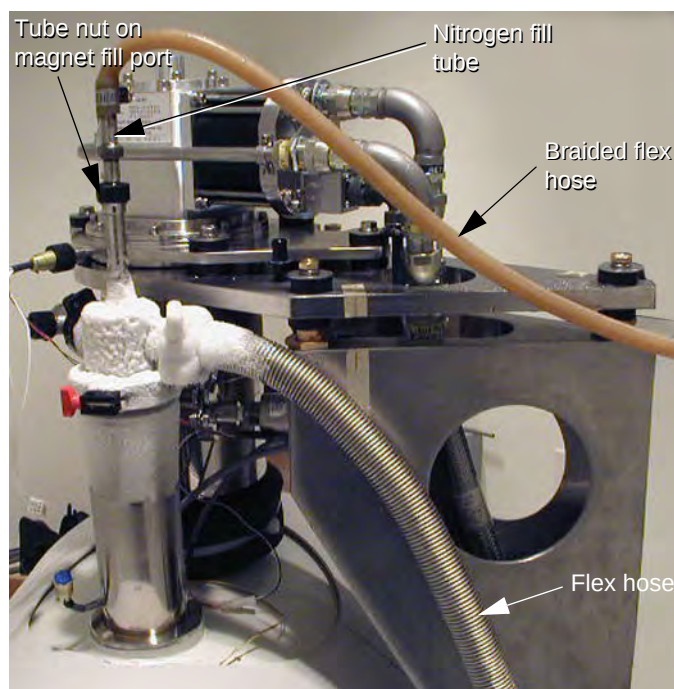


Figure 2-54: Double Bucket to Catch Nitrogen

8. Place the free end of the flex hose in a double bucket (e.g. a plastic trash can inside a metal container, see [Figure 2-54](#)) that will withstand the liquid nitrogen and that is large enough to hold 10 gallons (45 l) of liquid. Place a piece of insulating material such as hard foam under the bucket, to protect the floor.
9. Slowly open the dewar liquid port valve just enough to hear flow, then slowly open the valve all the way.
Liquid nitrogen entering the magnet produces an ear piercing resonance.
10. Adjust the valve to make the noise tolerable, while maintaining a reasonable fill rate.
11. The noise should stop or be reduced when liquid nitrogen accumulates in the bottom of the magnet.

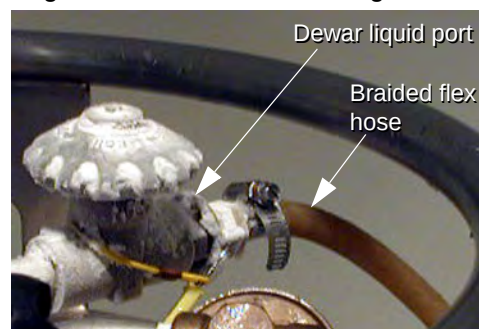


Figure 2-55: Fill Hose on Nitrogen Dewar

12. Connect a digital voltmeter (DVM) on the H and J pins of the Allen Bradley sensor magnet services connector. This connection reads about 420 Ω to 450 Ω when the magnet reaches liquid nitrogen temperatures.

The fill takes approximately one hour.

13. When liquid nitrogen runs out of the flex hose and into the bucket, close the dewar liquid port valve.
14. Let the set-up sit for at least four hours. If it stands longer than six hours, top off the nitrogen in the magnet before beginning the purge procedure.
15. Continue to record the values named in [Testing the MMU Sensors and Quench Button](#) on page 31 in the appropriate sections of the *Magnet Characterization Data Sheet* while the magnet is cooling.



Purging the Nitrogen

You must purge the magnet of all liquid nitrogen using helium gas before filling the magnet with liquid helium. Nitrogen remaining in the magnet freezes in liquid helium, compromising the helium's ability to cool the magnet.

WARNING: Pressurized gas bottles can be dangerous if they fall over. Always make sure that gas bottles are secured to a wall or a stable cart, or place the bottles horizontally on the floor.

WARNING: Asphyxiation hazard! The nitrogen purge/precool procedures release nitrogen gas, which will displace oxygen in the air and potentially cause asphyxiation. Ensure the following:

- The magnet room vent system is installed and functioning in conformance with site requirements.
- The magnet room door is propped open.
- A portable floor exhaust fan (>850 CFM or >24 m³/min) is installed and functioning.
- An oxygen monitor is installed and functioning in the magnet room in conformance with site requirements, or that service personnel use properly functioning oxygen monitors in the scan room.

WARNING: *Potential cold burn hazard! You may be exposed to cryogenics or contact cold objects during service procedures or a magnet quench. When performing any service procedure that uses cryogenics, wear protective clothing:*

- Long sleeve shirt
- Long pants
- Protective apron or jacket
- Non-absorbent gloves
- Glasses or goggles and a face shield

Figure 2-57: Nitrogen Purge Setup



1. Record the values of the link, the shield, the first stage, and the Allen Bradley switch on the magnet characterization data sheet, as described in [Characterizing the Magnet](#) on page 34.
2. Close the pressure regulator on the helium gas cylinder by turning the adjusting knob counter-clockwise.
3. Attach the gas transfer hose from the transfer kit to the regulator on the gas bottle.
4. Remove the flex hose from the service turret and attach the hose from the gas bottle in its place.
5. Disconnect the braided flex hose from the nitrogen dewar, and place the end of the hose in the double bucket used earlier. (See [step 8](#) on page 37.)

The braided flex hose is used as a vent tube in the purge process.

6. Open the helium bottle valve until you hear the gas at the vent tube.
7. Slowly turn the regulator knob clockwise until the regulator pressure gauge just begins to register (≈ 2 PSI).
8. Open the regulator valve until you hear the gas at the vent tube.

Within a few minutes, liquid should flow from the vent tube into the bucket.

If no liquid appears:

- a. Increase the pressure slightly.
If this does not produce liquid, the vent tube may not be seated correctly.
- b. Close the regulator valve and try reseating the vent tube, then repeat the process.

Approximately 2 gallons of liquid nitrogen will flow into the bucket before liquid stops coming out and only gas emerges. You may have to adjust the regulator to maintain a reasonable flow of gas through the magnet.

9. Allow the gas flow to continue for about 10 minutes to ensure that all the liquid nitrogen is purged.
10. During this time, monitor helium level by setting the MMU to constantly update the level reading.

This ensures that any liquid nitrogen caught in the helium probe's protective tube is vaporized.

11. Close the helium bottle valve, remove the braided stainless flex hose from the magnet, and remove the fill tube from the fill port.
12. Remove the tube nut, the brass washer, and the O-ring from the fill tube and set them aside. (See [Figure 2-53](#) on page 36.) Plug both orifices with tissue.

The removed parts are used in the procedure that follows.

Note: Begin the liquid helium fill within 10 minutes, to prevent excessive warming.

13. If possible, don't move the bucket of liquid nitrogen until all the nitrogen has evaporated. If you need to move the bucket of liquid, use extreme caution.
14. Record the values of the link temperature, the shield temperature, the first stage temperature, and the Allen Bradley temperature on the magnet characterization data sheet, as described in [Characterizing the Magnet](#) on page 34.

Filling With Liquid Helium

WARNING: *Asphyxiation hazard! The cooling procedure releases helium gas, which will displace oxygen in the air and potentially cause asphyxiation. Ensure the following:*

- The magnet room vent system is installed and functioning in conformance with site requirements.
- The magnet room door is propped open.
- A portable floor exhaust fan (>850 CFM or >24 m³/min) is installed and functioning.
- An oxygen monitor is installed and functioning in the magnet room in conformance with site requirements, or that service personnel use properly functioning oxygen monitors in the scan room.

WARNING: *Potential cold burn hazard! You may be exposed to cryogenics or contact cold objects during service procedures or a magnet quench. When performing any service procedure that uses cryogenics, wear protective clothing:*

- Long sleeve shirt
- Long pants
- Protective apron or jacket
- Non-absorbent gloves
- Glasses or goggles and a face shield

Protecting the Floor During the Helium Fill

During the helium fill, ice will form on the vent lines and fittings and liquid oxygen will drip. Prevent liquid oxygen from falling directly on the magnet body. Put isolating material such as plywood, styrofoam or several layers of cardboard on the floor below the vent lines, since the liquid oxygen is likely to crack the flooring.

Inserting the Fill Tube in the Helium Dewar

Most helium dewars have four valves, as shown in the photo at right. The valves are:

- The liquid valve
 - The vent valve
 - The pressure release valve
 - The pressure control circuit valve
1. Open the CasWin program from the Service Tools menu in Service mode. (See [step i](#) through [step j](#) on page 32 for instructions on opening CasWin.)
 2. Connect the MMU test fixture to the Magnet Services cable, and verify proper readings.
 3. Connect the Magnet Services cable to the MMU.

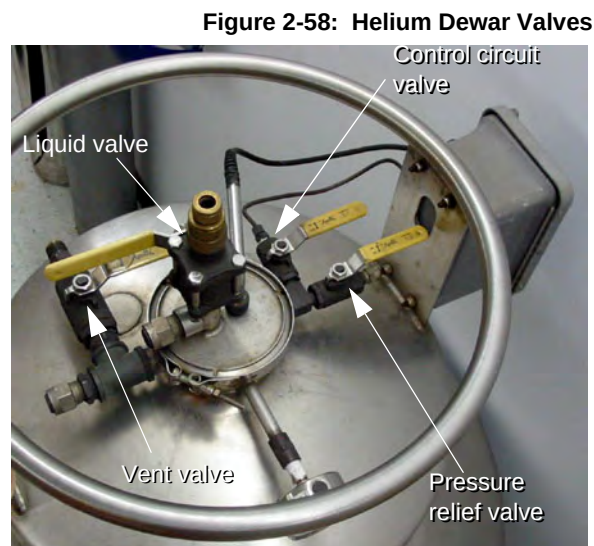


Figure 2-58: Helium Dewar Valves

4. Click the MMU1 tab in the CasWin window, then click the Helium Level button to check the helium level.
 5. Install the tube nut, brass washer, and O-ring (from [step 12](#) on page 38) on the fill tube.
 6. Loosen the brass tube nut on top of the liquid valve of the helium dewar.
 7. Find an area where the ceiling is high enough to clear the fill tube. If necessary, put the dewar in another room.
- If a high enough space cannot be found, insert the fill tube into the dewar in two parts.

Note: The long extension tube should only be used with 250-liter and 100-liter dewars. Use the short extension tube with 60-liter and 30-liter dewars.



Figure 2-59: Helium Level in CasWin

8. Partially open the dewar vent valve to relieve some of the pressure.
9. Open the dewar liquid valve, and begin to slowly insert the fill line into the dewar.

Too much boil-off through the liquid port can freeze the O-rings, making it difficult to seat them properly.

When inserting the fill line in two parts (because of low ceilings):

- a. Wrap the top of the extension tube with tape to prevent its falling inside the dewar.
- b. Thread the upper section into the extension tube, then remove the tape.

Caution: Using vise-grips or other types of clamps to prevent the fill line extension tube falling into the dewar is not recommended, because they can damage the extension tube. If you must use clamps, wrap the jaws with several layers of cloth or electrical tape, and use the smallest clamping force that will hold the clamp in place. Try not to crimp or deform the tube.

10. Close the dewar vent valve and continue to slowly slide the fill tube into the dewar until it hits bottom.

If the fill tube does not have a separator on the end, bring the line up half an inch (12 mm) when it hits bottom.

The pressure created by sliding the tube to the bottom of the dewar is enough to start the helium flowing.

Figure 2-60: Fill Tube Separator



Inserting the Fill Tube in the Magnet

1. Allow the fill tube to cool until the gas escaping from the free end of the tube forms a distinct plume. This means liquid is getting through the tube. When this happens, immediately insert the fill tube into the magnet until it bottoms out, then screw the fill tube into the magnet at least $\frac{3}{4}$ of a turn.
2. Pull up on the fill tube to verify that it is threaded in. Tighten the tube nut to clamp the O-ring tightly against the tube.

Filling the Magnet

Note: If liquid condensation forms on the fill line, the fill line has lost its vacuum insulation and is not flowing liquid helium. The fill line must be replaced. Stop the fill, remove the fill line, and close everything back up until a working fill line is obtained.

1. When using a dewar without a built-in pressure control circuit (for example, the 100-, 60-, or 30-liter dewar):
 - a. Purge the air from the hose of a regulated helium gas bottle (see [Purging the Gas Fill Assembly](#) on page 30).
 - b. Connect the helium fill hose to the dewar vent valve.
 - c. Partially open the dewar vent valve.
 - d. Adjust the regulator on the gas bottle to build up 3 to 4 PSI in the dewar.

Note: Many of the dewar pressure gauges do not work correctly, so you must develop a feel for the flow rate of the helium exhaust from the magnet.

- Too high a flow rate wastes helium.
- Too low a flow rate results in a slow fill.

2. Connect the pressure control transducer and open the pressure transducer valve, if one is mounted on the dewar. Switch the pressure control transducer to 4 PSI.

An indicator lights, to show that the unit is functioning and pressure is beginning to build in the dewar.

As the magnet begins to fill with liquid helium, the Allen Bradley sensor reading changes to about 1 K Ω when the liquid helium begins accumulating. Soon after, the MMU begins registering the liquid.

Note: Make sure there is sufficient flow of helium exiting the vent. If the flow is too small, raise the fill tube slightly in the service turret.

3. Monitor the helium level in CasWin, to verify that the fill is progressing correctly.

4. If the level stops rising, check the pressure in the dewar. If the pressure is OK, the fill tube may have an ice block. To remove ice blocks:
 - a. Remove the fill line from the dewar and the magnet. (See [Removing the Helium Fill Line](#) on page 41.)
 - b. Thaw the line.
 - c. Blow out the line with helium gas.
 - d. Start over, beginning with [Inserting the Fill Tube in the Helium Dewar](#) on page 39.

Note: If the transfer must be interrupted and the fill line removed, plug both ends of the fill line to prevent ice blocks.

5. When CasWin indicates that the helium level is 250 mm, turn on the compressor and coldhead.
6. Continue the fill until the boil-off coming out of the magnet relief valve port develops a distinct plume about 6 inches (150 mm) long.

The helium level displayed in CasWin should read approximately 700.

Note: If you turn off the pressure transducer when the level reaches ≈ 625 mm, enough pressure should remain in the dewar to complete the fill. Doing this reduces liquid helium spillage when removing the fill tube.

7. When the fill is complete, turn off the pressure control transducer and unplug it.

Removing the Helium Fill Line

1. Close the valve to the pressure control circuit if it exists. If the dewar does not have a built in pressure control circuit, close the vent valve used for the helium gas input.
2. Close the helium gas bottle valve.
3. Remove the gas tube from the liquid helium dewar vent valve.
4. Open the dewar vent valve to bleed off the pressure in the dewar.
5. When the pressure in the dewar is relieved, remove the fill tube from the magnet.
6. Plug the fill port and quench valve port with tissue.
7. Remove the fill tube from the dewar, being careful not to bend the tube.

If you had to insert the fill line in two parts, when removing the line from the dewar, raise the fill line enough so that the threaded section is above the liquid fill valve and hold the tube in place by tightening the tube nut.

8. Close the dewar liquid valve and vent valve, and open the pressure relief valve.
9. With gloves on, install the quench vent, as described in [Installing the Quench Vent and Hose](#) on page 27. Do not install the flex hose yet.
10. Assemble the service turret plug with the tube nut and O-ring. Install the plug on the fill port and tighten it securely.
11. Connect the flex hose to the quench vent. Be aware that very cold gases are typically expelled from the quench vent about 10 minutes after a helium fill.
12. Record the values of the link temperature, the shield temperature, the first stage temperature, and the Allen Bradley temperature (described in [Characterizing the Magnet](#) on page 34).
13. If shield temperatures are still high (≈ 200 K is typical at this stage), the helium boil-off remains high for 6 to 12 hours. Allow the magnet to sit idle, with the coldhead running, until the link and shield temperatures have come down. This may take overnight.
14. When the temperatures have stabilized, remove the quench vent, top off the helium, re-install the quench vent and ramp the magnet.
15. An hour or more after the helium fill, verify that the collar nut is tight on the service turret. The cooling down of the o-ring and hardware may have caused the nut to become loose.

Ramping the Magnet

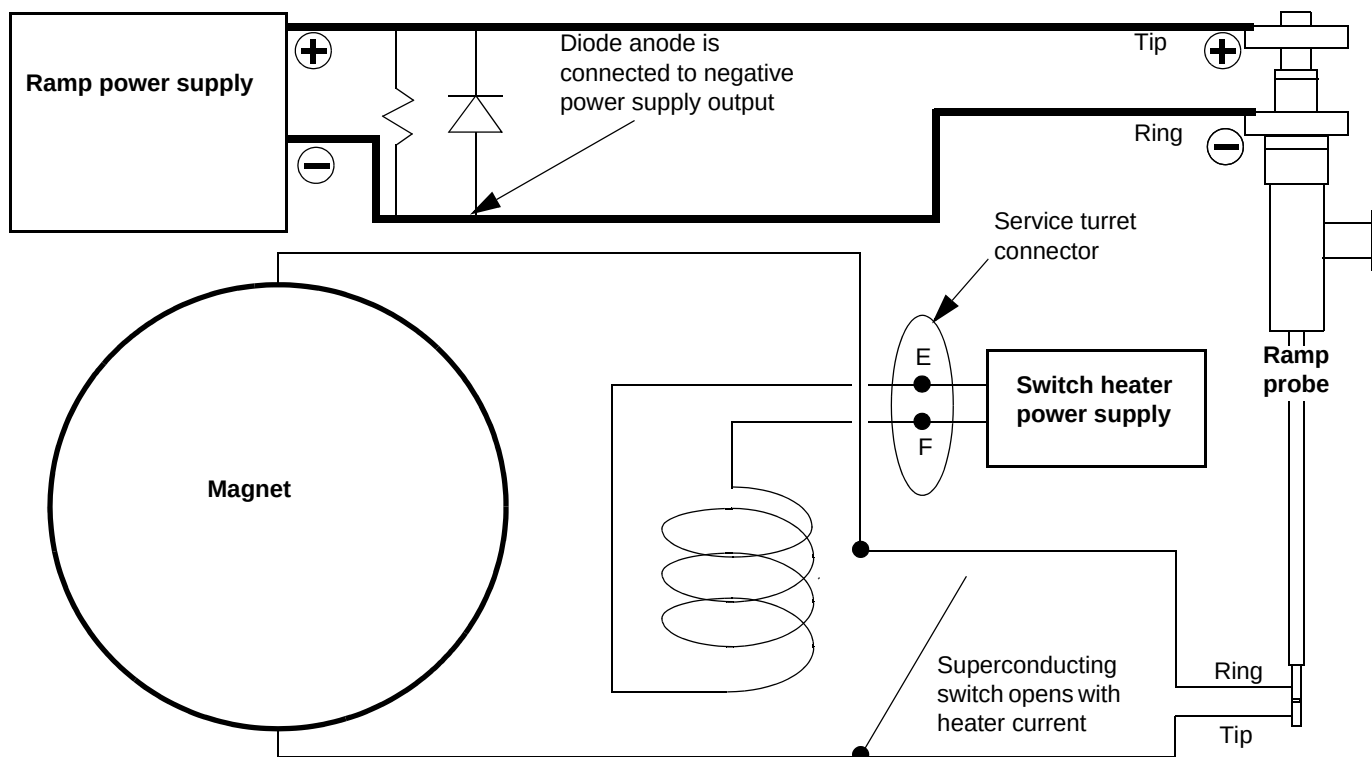
This procedure brings the magnet to field, the final step before configuring the system and generating images.

Setting Up the Equipment

Determine which ramp power supply you have in your kit: XFR6-200 or XG6-220. You can identify them by the name on the front of the power supply. If you are using the XG6-220 power supply, you need to use the Z-Brackets when making the power supply connections with the supplied hardware provided in the kit. After setting up the equipment, see [Energizing \(“Ramping”\) the Magnet Using the XFR6-200 Power Supply](#) on page 44 for instructions on ramping the magnet using the XFR6-200 power supply. See [Energizing \(Ramping\) the Magnet Using the XG6-220 Power Supply](#) on page 45 for instructions on ramping the magnet using the XG6-220 power supply.

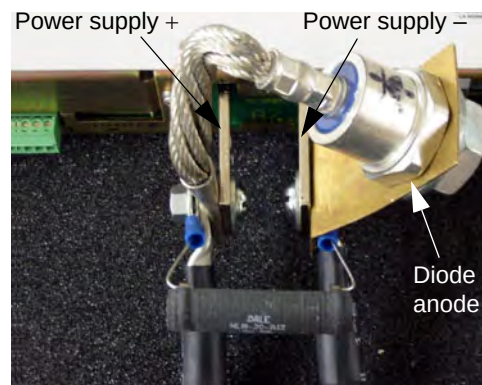
1. Open the ramp tool kit (see [page 84](#)), and remove its contents.
2. Set up the magnet power supply as shown in the figures below.

Figure 2-61: Magnet Ramp Setup



Note: The diode protects the power supply from damage in case of a quench during ramp-up.

Figure 2-62: Ramp Power Supply Connection





Figures: Diode Bracket with Z-Brackets for XG6-220, Z-Brackets attached to Power Supply, Diode Bracket attached using Z-Bracket

Figure 2-63: Magnetometer

Refer to the photo at right when setting up the magnetometer.

3. Set the magnetometer to measure frequency during the ramp.
4. Push in the **AUT** button on the magnetometer.



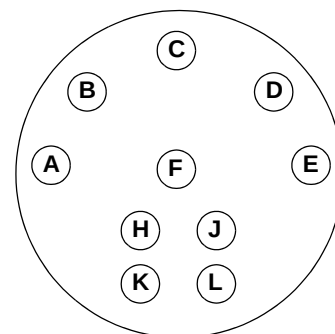
Figure 2-64: Magnetometer Probe



5. Connect the magnetometer probe to the rear of the magnetometer at the Power and RF OUT terminals.
6. Carefully position the tip of the magnetometer probe at the iso-center of the magnet (center the probe tip along the axis of the coil.) Use foam or similar material to raise the probe to approximately the vertical center of the magnet.

Figure 2-65: Service Turret Connector Pins

7. Use the break-out box, banana jacks E (switch heater) and F(common), to connect the switch heater power supply to the service turret. If you do not have a break-out-box, use BNC clip-on cables from the output connectors of the switch heater power supply to the service turret connector pins **E** (switch heater 1) and **F** (common).
8. Turn on the switch heater power supply and adjust the voltage control to supply 70 mA. Turn off the power supply.
9. Install and power up a portable floor exhaust fan (>850 CFM or >24 m³/min).
10. Remove the top section of the service turret and remove the tube nut, brass washer, and O-ring.
11. Install the tube nut, brass washer, and O-ring on the ramp probe.
12. Look down the service turret to check for ice on the fill port (use helium gas to melt any ice found) and insert the ramp probe in the magnet.



Energizing (“Ramping”) the Magnet Using the XFR6-200 Power Supply

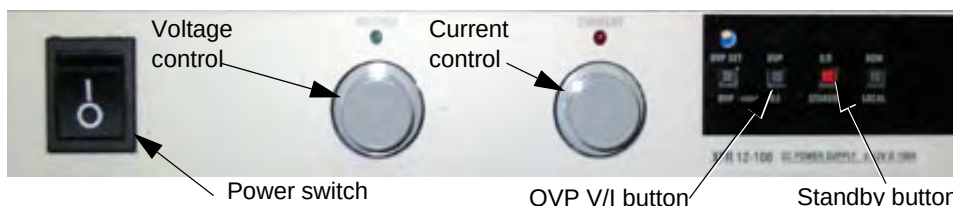
This section describes how to energize the magnet using the XFR6-200 power supply.

WARNING: *The GE MR system creates a strong magnetic field, which can turn metal objects into missiles. To prevent personal injury and equipment damage, thoroughly search the Exam room, and remove all tools and other objects that might be affected by the magnetic field, before energizing the magnet.*

WARNING: *A quench can happen at any time during a magnet ramp. During a quench extremely cold helium gas may flow into the scan room. Do not enter the scan room while the magnet is being ramped.*

Figure 2-66: Ramp Power Supply

Refer to the photo at right for ramp power supply setup and control.



1. Before turning on the ramp power supply, press the red **Standby** button on the front of the supply, to place it in standby.
2. Turn the **Voltage control** knob counter-clockwise all the way.
3. Turn on the **ramp power supply**, and while holding in the **OVP/VI** button, adjust the **Current control** to produce 170 A.
4. Turn on the **switch heater power supply**, verify that the heater power supply display still reads about 70 mA, and wait about 15 seconds.
5. Begin ramping the magnet by pressing the red **Standby** button on the ramp power supply to take it out of standby.
6. Adjust the **Voltage control** knob to produce **2.5 volts**.

The displayed current should begin to increase slowly.

- If the current increases rapidly, the probe is not seated properly. It is most likely shorted to the magnet internally. Reseat the probe. Ice may have built up in the service turret. If so, blow out the turret with helium gas.
- If there is no current, the probe is not seated far enough into the magnet. Insert the probe farther into the magnet.

7. Monitor the current displayed on the **ramp power supply**.

Because of the voltage drop across the ramp leads, the rate of change of the ramp current is slow. Continue increasing the voltage slightly (approximately 0.5 V at 50 A intervals) to maintain a slow and steady current rate.

8. When 50 A is reached, adjust the voltage to **3.0 volts**.
9. At approximately 60 A, press the **Search** button on the magnetometer to start an auto-search.

The magnetometer should lock in.

10. When 100 A is reached, adjust the voltage to **3.5 volts**.
11. When 150 A is reached, adjust the voltage to **4.0 volts**.
12. When the ramp supply display reaches 170 A, the LED directly above the current knob lights. Check the frequency display on the magnetometer. Adjust the **Current control** on the ramp supply to produce a frequency between 42.610 MHz and 42.628 MHz if this is a new installation.
13. Adjust the current knob in small increments, to allow the frequency to settle after each adjustment. Large or rapid adjustments will overshoot the target amperage, and the time spent recovering may cause the magnet to quench.

Note: You do not have a lot of time to park the 1.0T magnet, like you do with the 1.5T magnet. You must reach your target frequency for the 1.0T magnet as soon as possible. If you do not, it may quench.

14. If you are recovering from a quench or a coldhead replacement, target the frequency before the quench or coldhead replacement. The specification for the center frequency is **42.575 MHZ** to **42.625 MHZ**. Allow some room for shimming. The typical park current for a 1.0T magnet will be approximately **172 amps** in most cases.
15. When the target frequency is reached, shut off the switch heater power supply and wait for about **15 seconds**.

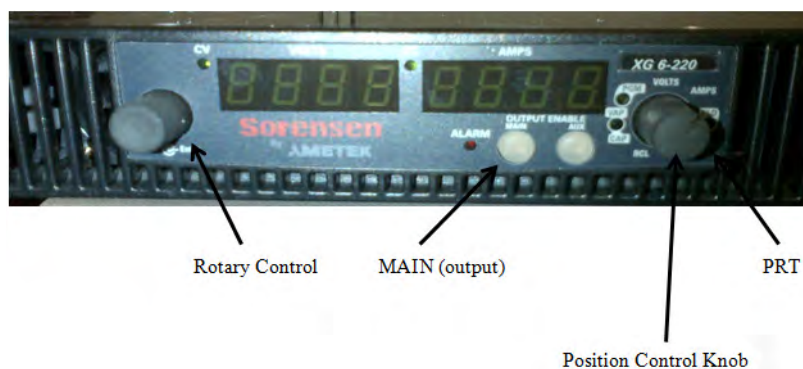
16. Begin to ramp down the ramp power supply by turning the **Current control** knob slowly, and verify that the frequency does not drop quickly.
 - If the frequency is holding, continue to turn the current down. When the current reaches zero, turn off the ramp power supply.
 - If the frequency drops quickly and continues to drop steadily, the switch heater has not cooled enough. Quickly turn the switch heater power supply on, and adjust the ramp power supply. If this does not work, shut down the switch heater power supply, or the magnet will quench.

The safest way out of this situation is to ramp down the magnet and repeat this procedure starting at [step 1](#).
17. Remove the ramp probe from the magnet, and install the top section of the service turret, the tube nut, the washer, and the O-Ring.
18. Record the final **ramp current value** and the **frequency** in the *Shim and Park Values* table of the Installation data sheet.

Energizing (Ramping) the Magnet Using the XG6-220 Power Supply

This section describes how to energize the magnet using the XG6-220 power supply. If you have never used the XG6-220 power supply, we recommended you review all the steps in the procedure before you begin.

Figure 2-67: XG6-220 Power Supply



1. Turn on the power supply and ensure that **MAIN** is in the off position (light is off).
2. Set the OVP setting by turning the **Position Control** knob to **PRT**. You will see **Pr0 OVP** displayed,
3. Press the **Rotary Control** knob. You will now see **OVP** and a voltage value displayed.
4. Set the OVP to **4.0** volts. If it requires adjusting, turn the **Rotary Control** knob to achieve the desired voltage. If you need to **fine** adjust, press the knob one more time and you will see **OVPF** displayed. Once the voltage is set correctly, press the **Rotary Control** to set the voltage. (The voltage may already be set for you from the previous user.)
5. Turn the **Position Control** to **amps**.
6. Turn the **Rotary Control** knob to **170 amps**. Once 170 amps is reached, press in the **Rotary Control** knob to set the current. If you need to fine tune in order to reach 170 amps do the following:
 - a. Slowly press the **Rotary Control** knob in once. Pause 1 to 2 seconds.
 - b. Slowly press the **Rotary Control** knob in two (2) more times. You will see **CrPC** displayed. Pause 1 to 2 seconds each time you press the knob.
 - c. Slowly press the **Rotary Control** knob in one more time (a fourth time). You will see **FnPC** displayed.
 - d. Adjust the **Rotary Control** knob to reach 170 amps and press the **Rotary Control** knob in to set the current to 170 amps. If you do not see **CrPC** or **FnPC** after pressing the knob 4 times, you may have pressed the knob too fast. Continue to slowly press it and you will see **CrPC** or **FnPC** displayed.

Note: The FnPC mode is temporary and the supply will switch back to normal mode after a few minutes. Make sure that you are still in FnPC mode any time you are making adjustments.

Note: Any time an adjustment is made with the current setting throughout this process you **MUST** press the **Rotary Control** knob to set the new value.
7. Turn on the switch heater power supply and wait 15 seconds.
8. Press the **MAIN** output enable. The light will turn on.
9. Turn the **Position Control** to **volts** to slowly adjust the voltage to approximately 2.5 volts. The displayed current will begin to increase. The displayed current should begin to increase slowly.

- If the current increases very rapidly, the probe is either not seated properly or not enough time was allowed for the switch heater to turn on. If the switch heater is on and the current still increases rapidly, the probe is most likely shorted to the magnet internally. Reseat the probe. Ice may have built up in the service turret. If so, blow out the turret with helium gas.
 - If there is no current, the probe is not seated far enough into the magnet. Insert the probe farther into the magnet.
10. The magnet will begin to ramp. Monitor the current displayed on the ramp power supply.
Note: Because of the voltage drop across the ramp leads, the rate of change of the ramp current is slow. You will be increasing the voltage slightly (approximately 0.5 V at 50 A intervals) to maintain a slow and steady current rate.
 11. When **50 A** is reached, adjust the voltage to **3.0** volts.
 12. At approximately **60 A**, press the **Search** button on the Teslameter to start an auto-search. The Teslameter should scan and lock in.
 13. When **100 A** is reached adjust the voltage to **3.5** volts.
 14. When **150 A** is reached adjust the voltage to **4.0** volts.
 15. When the ramp supply reaches **170 A**, the CC LED will light. Note the frequency on the Teslameter.
 16. Turn the **Position Control** to **AMPS**.
 17. Press the **Rotary Control** knob and set for “fine adjustment.”(Refer to [step 4.](#)) When you see **FnPC** displayed, slowly adjust the current towards the direction of the target frequency. Adjust in small increments and press the **Rotary Control** knob each time to set the new current.
 18. Adjust the current knob in small increments, to allow the frequency to settle after each adjustment. Large or rapid adjustments will overshoot the target amperage, and the time spent recovering may cause the magnet to quench.
Note: You do not have a lot of time to park the 1.0T magnet, like you do with the 1.5T magnet. You must reach your target frequency for the 1.0 magnet as soon as possible. If you do not, it may quench.
 19. If you are recovering from a quench or a coldhead replacement, target the frequency before the quench or coldhead replacement. The specification for the center frequency is **42.575 MHZ** to **42.625 MHZ**. Allow some room for shimming. The typical park current for a 1.0T magnet will be approximately **172** amps in most cases.
 20. When the target frequency is reached, shut off the switch heater power supply and wait for about **15 seconds**.
 21. Begin to ramp down the ramp power supply by turning the **Rotary Control** knob very slowly, and verify that the frequency does not drop quickly.
 - If the frequency is holding, continue to turn the current down. (Now you can turn it down faster). Once the current reaches zero amps, turn the **Position Control** to **Volts**, adjust the voltage to zero, and turn off the ramp power supply.
 - If the frequency drops quickly and continues to drop steadily, the switch heater has not cooled enough. Quickly turn the switch heater power supply on, and adjust the ramp power supply to re-park the magnet. Try turning off the switch heater again, this time waiting longer before dialing down the current.
 22. Remove the **ramp probe** from the magnet by pulling it straight up. Avoid twisting the ramp probe. Install the **top hat** assembly.
 23. Record the final **ramp current value** and the **frequency** in the *Shim and Park Values* table of the Installation data sheet.

Ramping Down the Magnet Using the XFR6-200 Power Supply

The following procedure ramps down the magnet using the XFR6-200 power supply. This can be necessary if, for example, the system must be moved, or when replacing the coldhead.

1. Verify that the magnet **helium level** is above 450. If it is not, top it off following the procedures in [Filling With Liquid Helium](#) on page 39.
2. Set up the **ramp power supply** as shown in [Figure 2-66](#) on page 44.
3. Turn on the **switch heater power supply** with no output cables connected, turn the current control to approximately the 9 o'clock position and adjust the voltage control to supply approximately 7.0 volts. *Turn off the switch heater power supply.*
4. Connect the switch heater cable to the power supply and use the **break-out box** or **clip-on leads** to connect to the service turret connector pins **E** (switch heater 1) and **F** (common). *Do not turn on the switch heater power supply.*
5. Remove the top assembly from the **service turret**, and remove the **tube nut**, **brass washer**, and **O-ring**.
6. Install the tube nut, brass washer, and O-ring on the **ramp probe**, and insert the probe in the magnet.
7. Set the ramp power supply to **Standby**, and turn the power supply on.
8. Turn on the **switch heater power supply**, and verify that it reads approximately 70 mA. Adjust the voltage control if necessary.

This opens the magnet circuit.

The ramp power supply now acts as a current monitor. When you turn on the switch heater power supply, you should see approximately 170 A on the ramp power supply display.

Note: If the current does not appear immediately on the ramp power supply display, the probe is not seated properly. Shut off the switch heater power supply immediately and reset the probe. Running the switch heater with the probe not seated will quench the magnet.

9. Continue to monitor the **current** on the supply.

As the magnet ramps down, the current displayed on the power supply decreases.

10. When the current displayed on the power supply reaches **0**, turn off the **ramp power supply** and the **switch heater power supply**.
11. Disconnect the **power supplies**, remove the **probe**, and reassemble the **service turret**.

Ramping Down the Magnet Using the XG6-220 Power Supply

The following procedure ramps down the magnet using the XG6-220 power supply.

1. Verify that the magnet **helium level** is above **450 mm**. If it is not, top it off following the procedures in [Filling With Liquid Helium](#) on page 39.
2. Set up the **ramp power supply** as shown in [Figure 2-67](#) on page 45.
3. Remove the top hat assembly from the **service turret**, and replace it with a tissue. Remove the **tube nut**, **brass washer**, and **O-ring**.
4. Turn on the **switch heater power supply** with no output cables connected. Turn the current control to approximately the 9 o'clock position and adjust the voltage to supply approximately 7.0 volts. Turn off the switch heater power supply.
5. Connect the switch heater cable to the power supply and use the **break-out box** or **clip-on leads** to connect to the service turret connector pins E (switch heater I) and F (common). Do not turn on the switch heater power supply.
6. Install the tube nut, brass washer, and O-ring on the **ramp probe**, and insert the probe in the magnet.

NOTICE

If you have an extra set of service turret hardware (available in the 5000-7725 Consumables Kit) to use with the ramp probe, that is the preferred method in order to avoid keeping the service port open longer than necessary.

7. Turn on the Power Supply.
8. Turn the **Position Control** knob to **AMPS**.
9. Turn the **Rotary Control** to **1-3 AMPS** and press the **Rotary Control** knob in to set the current.
10. Turn the **Position Control** knob to **VOLTS**. Set the voltage to zero.

Note: Setting the current to **1-3 AMPS** allows the power supply to continuously report the current level during the ramp down. If you do not set the current, the magnet will still ramp down but you will have to switch between AMPS and VOLTS on the Position Control knob in order to update and view the status of the current.

11. Press **MAIN** to turn it off (the light will turn off if it is turned on).
12. Turn on the **switch heater power supply**, and verify that it reads approximately 70 mA. Adjust the voltage control if necessary.
This opens the magnet circuit.

The ramp power supply now acts as a current monitor. When you turn on the switch power supply, you should see approximately 170 A on the ramp power supply display.

Note: If the current does not appear immediately on the ramp power supply display, the probe is not seated properly. Shut off the switch heater power supply immediately and reset the probe. Running the switch heater with the probe not seated will quench the magnet.

13. Continue to monitor the current on the supply. As the magnet ramps down, the current displayed on the power supply decreases.
14. When the current displayed on the power supply reaches the setting in [step 8](#), with the position control knob set to amps, adjust current for zero amps. Press in the **Rotary Control** knob. Continue to monitor the current until it reads **0 AMPS**.
15. Turn off the **ramp power supply** and the **switch heater power supply**.
16. Disconnect the **power supplies**, remove the **probe**, and reassemble the **service turret**.

3 System Configuration and Test

Chapter Contents

This chapter includes instructions for testing and characterizing an MR imaging system (Part #2000-0000) at a customer site, using the *System Installation Data Sheet* (DOC0922205). The completed data sheet can be used as one criterion for system turnover to the customer.

The chapter includes the following topics:

- [Hardware Setup](#) on page 50
- [Software and PC Configuration](#) on page 51
- [Installing Service Software](#) on page 57
- [RF Loopback Testing](#) on page 58
- [Gradient Amplifier Calibration](#) on page 61
- [Shim \(First Pass\)](#) on page 62
- [Eddy Current Compensation](#) on page 63
- [Shim \(Second Pass\)](#) on page 63
- [Characterizing the Center Frequency](#) on page 64
- [Magnet Monitoring Unit](#) on page 64
- [Magnet Pressure Regulator](#) on page 67
- [Characterizing the Field Stability](#) on page 68
- [Characterizing the 180 mm RF Coil](#) on page 69
- [Characterizing the 123 mm RF Coil](#) on page 70
- [Characterizing Optional RF Coils](#) on page 71
- [Characterizing the Helium Compressor](#) on page 72
- [Miscellaneous Tests](#) on page 72
- [Characterizing Image Quality](#) on page 74
- [Modem Dial-In](#) on page 75
- [Image Archive](#) on page 75
- [DICOM Transfer](#) on page 76
- [Configuration Backup](#) on page 76
- [Removing Service Software](#) on page 77

Test Requirements

The procedures below must be performed in the order in which they are listed.

Test Equipment

These procedures require the Service CD and the calibration fixture tool kit (see [page 84](#)).

Other required test equipment varies with the procedure or test. Required items are listed in each procedure.

Test Conditions

The tests must be performed after the system is in the final clinical configuration, and after the site has met the pre-installation requirements listed in the *Site Planning Guide* (DOC0922291).

Visual and Mechanical Inspection

Some system components require visual or mechanical inspection. When applicable, these inspections are called out for each component.

Procedures

Preparing the Installation Data Sheet

You must enter data generated during these procedures in the *System Installation Data Sheet*. The data sheet is a Microsoft Excel® template on the Service CD.

Note: The preferred method for entering data in the data sheet is to type it into the appropriate cells in a new copy of the data sheet on your laptop computer. If this is not practical, you can print the blank data sheet, and write the data on the printout by hand. Only the Excel version is accepted by ONI, so handwritten data must still be entered in the Excel file.

To enter data into the data sheet on your computer:

1. Open the data sheet Excel template (DOC0922205_OrthOne_Install_Datasheet.xls).
2. Save the data sheet as a new file on your computer hard drive, with the site name in the file name.
Keep the new data sheet file open.

Figure 3-1: Switching Views With Alt-Tab



3. When you reach a step in these procedures that instructs you to enter data in the data sheet, press and hold the **Alt** key on your laptop keyboard, then press the **Tab** key. If necessary, press the Tab key again (while holding down the Alt key), until the Excel icon is selected.
4. Release the **Alt** key.
Your computer displays the new version of the Installation data sheet.
5. When you have finished entering the data and want to return to the Installation Guide, press **Alt-Tab** again.
Your computer displays this document.

Hardware Setup

Magnet Installation

1. Record the serial number of the gradient coil in the *Gradient Amp Calibration* table of the Installation data sheet.
2. Assemble the system and ramp the magnet to field, as described in [Chapter 2, System Installation](#).

Figure 3-2: Gradient Amp Calibration Table

Gradient Amp Calibration						
Gradient Coil Serial #						
Parameter	Units	Min	Max	Measured	P / F	Comments
B0 gain	*	*	*		*	
Grad_Gain_X	*	[17,167]	[18973]			
Grad_Gain_Y	*	[17,167]	[18973]			
Grad_Gain_Z	*	A: [17,140] B: [18,383]	A: [19,717] B: [20,317]			

Figure 3-3: Shim and Park Values Table

Shim and Park Values						
Parameter	Units	Target	Max	Measured	P / F	Comments
Magnet Park Frequency (before shimming)	MHz	*	*		*	
Magnet Ramp Current (at park frequency)	Amps	*	*		*	
DiaXY=150, DiaZ=120 (shim ellipsoid setting & val)	ppm	15	20			
DiaXY=150, DiaZ=120 (ref. ellipsoid value)	ppm	*	*		*	
DiaXY=100, DiaZ=80 (shim ellipsoid setting & val)	ppm	*	*		*	
DiaXY=99.1, DiaZ=80 (ref. ellipsoid value)	ppm	2.5	3			

3. Record the frequency and the final current required when the magnet was parked, in the *Shim and Park Values* table of the Installation data sheet.

Note: If no signal is available after ramping the magnet, the I and Q cables may be cross-connected. Swapping the I and Q cables at the RF front end can correct this condition.

System Serial Number

1. Label the system as described in the *System Safety Labels Process Sheet* (2000-0601). After completing the labeling, mark the *Miscellaneous Checks* table of the Installation data sheet.

Figure 3-4: Miscellaneous Checks Table

Miscellaneous Checks		
Function	Check	Comments
System Labeling	☐	
MRI Warning Signs	☐	
Laser Alignment Light	☐	
Patient Chair	☐	
Magnet Covers	☐	
Coil ID	☐	
Room RFI	☐	
Power Distribution Unit (PDU)	☐	
Quench Button	☐	
Safety Grounds	☐	
Equipment room temp	☐	
Magnet room temp	☐	
Configuration Backup	☐	

2. In the sysconfig.mdb file, update the Scanner_Serial_Num parameter value, and record the value in the *Site Information* table of the Installation data sheet.

The serial number is on the system label on the rear magnet cover.

3. In the sysconfig.mdb file, update the Scanner_Model field to say *OrthOne* or *MSK Extreme* as appropriate.

Figure 3-5: Site Information Table

Site Information (data taken from sysconfig.mdb file, must match final version exactly)		
Item	Value	Comments
Daily_QA_RX_Gain_Lower		
Daily_QA_RX_Gain_Upper		
Daily_QA_TX_Gain_Lower		
Daily_QA_TX_Gain_Upper		
RfAmp_Trip_Pt_Slope		
Institution Name		
Institution Address		
RX_DAC_Slope		
RX_DAC_Start		
Scanner_Serial_Num		
Station		
TX_DAC_Slope		
TX_DAC_Start		
UI_Date_Format		

MRI Warning Signs

Verify that the MRI warning signs are hung properly. Mark the *Miscellaneous Checks* table of the Installation data sheet.

Software and PC Configuration

This section describes how to configure the host PC. The system configuration must be updated to reflect the site and the system being installed. The software must also be configured to allow archiving, DICOM transfer, and printing film.

Software Configuration

The electronics cabinet is supplied with a permanent MAC address-based Silhouette® license installed. In this configuration, software reinstallation is not necessary.

Network Configuration

This procedure configures the host computer to communicate over the site network. The values used are provided by the site network administrator, and must be entered in the *Network Configuration* table of the Installation Date Sheet.

1. Power up the host computer and log in as **administrator** in order for the configurations in the following steps to take effect.

2. Click  Start, then click Control Panel.
The Control Panel window opens.

Figure 3-6: Opening Control Panel Window



3. Double-click Network and Internet Connections.
The Network and Internet Connections window opens.

Figure 3-7: Opening Network Control Panel



4. Click Network Connections.
The Network Connection Options window opens.

Figure 3-8: Network and Internet Connections Window



5. Click Local Area Connection.
The Local Area Connection Status window opens.

Note: If the customer’s data cable is connected to the host computer, the Local Area Connection icon is labeled Enabled, as shown in

Figure 3-9: Network Connection Options Window

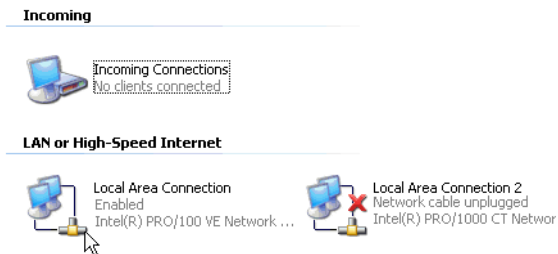


Figure-3-9. If the cable is disconnected, the icon is labeled Network cable unplugged, with a red X (like the Local Area Connection 2 in the figure).

6. Click Properties.
- The Local Area Connection Properties window opens.

Figure 3-10: Local Area Connection Status Window

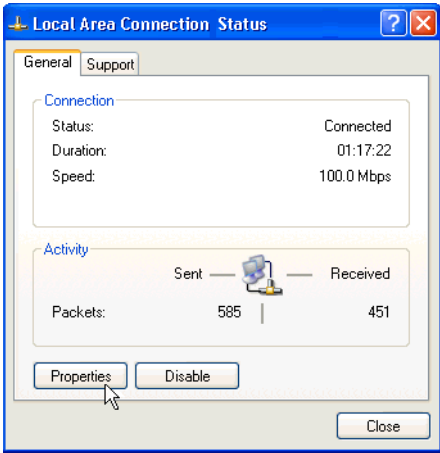


Figure 3-11: Local Area Connection Properties Window

7. Select Internet Protocol (TCP/IP), then click Properties.
- The Internet Protocol (TCP/IP) Properties window opens.

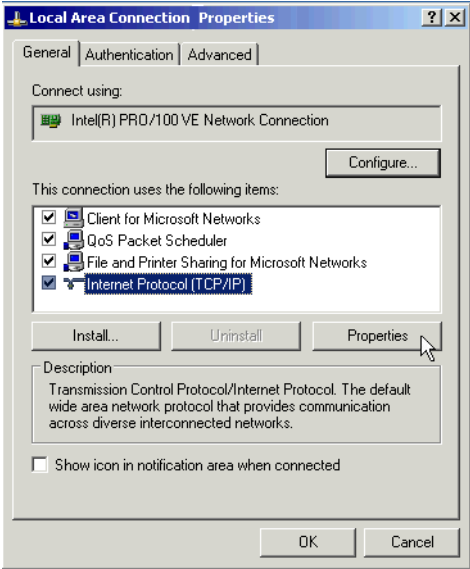
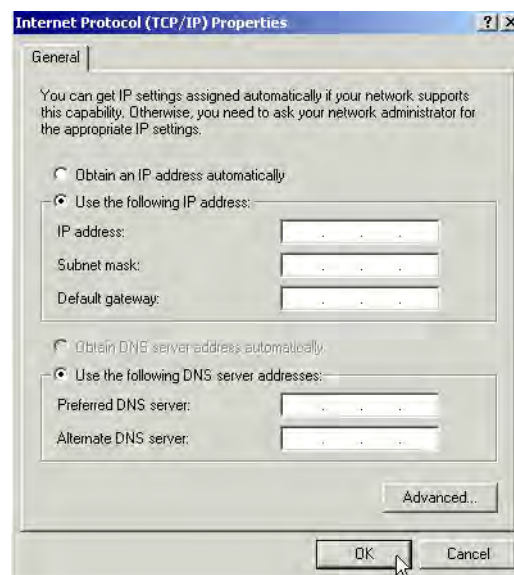


Figure 3-12: Internet Protocol (TCP/IP) Properties Window

8. Select Use the following IP Address:.
 9. In the IP Address:, Subnet Mask:, and Default Gateway: fields, enter the values provided by the site network administrator.
 10. Click OK.
- The Internet Protocol (TCP/IP) Properties window closes.
11. Close all open windows.



Assigning the Computer Name

Note: The default computer name is ExtremityMRI

- If the site administrator specifies another name, continue with this procedure.
- If the default is acceptable, skip to [Configurations](#) on page 56.

Figure 3-13: Control Panel Window

1. Click Start, then select Control Panel.

The Control Panel window opens.

Pick a category



2. Double-click Performance and Maintenance.

The Performance and Maintenance window opens.

Figure 3-14: Opening System Control Panel



3. Double-click System.

The System Properties window opens.

4. Click the Computer Name tab, and click Change....
The Computer Name Changes window opens.
- Figure 3-15: System Properties Window



5. Edit the computer name, if you want to change it from ExtremityMRI.
6. Select either Workgroup: or Domain:, and enter the workgroup or domain name provided by the site network administrator.
In Figure-3-16, Workgroup is selected as an example.
7. Click OK.
The Computer Name Changes window closes.
8. Click OK in the message boxes that open.
9. When the System Settings Change dialog box opens, click Yes.
The message box closes, and the host computer restarts.
- Figure 3-16: Computer Name Changes Window



10. Enter the **computer name** and the **workgroup name** or **domain name** in the *Network Configuration* table.
- Figure 3-17: Network Configuration Table

Network Configuration		
Item	Value	Comments
System Phone Number		
Host Computer's Network Name		
Host Computer's IP Address		
Domain Name (if any)		
Subnet Mask Value		
Gateway Address (if any)		

Configurations

1. In the *Network Configuration* table of the Installation data sheet, record the computer **network name**, **IP address**, **domain name**, **subnet mask**, and **gateway address** used during software installation. (These configurations are supplied by the site network administrator.)

Figure 3-18: Site Information Table

Site Information (data taken from sysconfig.mdb file, must match final version exactly)		
Item	Value	Comments
Daily_QA_RX_Gain_Lower		
Daily_QA_RX_Gain_Upper		
Daily_QA_TX_Gain_Lower		
Daily_QA_TX_Gain_Upper		
RfAmp_Trip_Pt_Slope		
Institution Name		
Institution Address		
RX_DAC_Slope		
RX_DAC_Start		
Scanner_Serial_Num		
Station		
TX_DAC_Slope		
TX_DAC_Start		

2. In the sysconfig.mdb file, update the following parameters to the values specified by the site network administrator:

- Institution Name
- Institution Address
- Station

Record these values in the *Site Information* table of the Installation data sheet.

3. In the sysconfig.mdb file, update the following parameter values with the values recorded on the back of the Exciter/Receiver, and record the values in the *Site Information* table of the Installation data sheet:

- RX_DAC_Slope
- RX_DAC_Start
- TX_DAC_Slope
- TX_DAC_Start

Note: These may be labeled as RX Slope, RX Offset, TX Slope, and TX Offset, respectively.

4. In the sysconfig.mdb file, update the RfAmp_Trip_Pt_Offset and the RfAmp_Trip_Pt_Slope parameters to the values recorded on the back of the RF amplifier, and record the value in the *Site Information* table of the Installation data sheet.

Note: For a CPC RF amplifier, RfAmp_Trip_Pt_Offset is always 0 and may not be written on the label.

Figure 3-19: RX_DAC and TX_DAC Labels



Figure 3-20: Software / Computer Table

5. Check the **SW install** box in the *Software / Computer* table of the Installation data sheet.

Software / Computer		
Software Version #		
Function	Check	Comments
ONI SW install	☐	
Modem dial-in test	☐	
Image Archive test	☐	
Monitor check	☐	

Changing the Date Format to European Mode

Systems are configured, by default, to use the “American” date format of month/day/year (mm/dd/yyyy). If the system is installed in a site using the “European” format (dd/mm/yyyy), you must change the format.

To change the date format in v.2.10:

1. Use Windows Explorer to navigate to the SERVICE TOOLS directory on the Service CD.
(GE Service Documentation\Tools, SW, Drivers, etc\2.10\SERVICE TOOLS)

Figure 3-21 Date Mode Batch Files

2. Double-click Set European Date Format.bat to set the date format to international mode.



Note: To turn international date mode off, Double-click Set American Date Format.bat.

Figure 3-22: Network Configuration Table

Network Configuration		
Item	Value	Comments
System Phone Number		
Host Computer's Network Name		
Host Computer's IP Address		
Domain Name (if any)		
Subnet Mask Value		
Gateway Address (if any)		

Modem Dial-In

1. With the scanner PC powered on, have someone log on remotely to the scanner.
2. Have them verify that files can be copied from the scanner system and placed onto the scanner system.
3. Record the telephone number used to connect to the modem in the *Network Configuration* table of the Installation data sheet.

Figure 3-23: Software / Computer Table

Software / Computer		
Software Version #		
Function	Check	Comments
ONI SW install	☐	
Modem dial-in test	☐	
Image Archive test	☐	
Monitor check	☐	

4. Check the Modem dialin box in the *Software / Computer* table of the Installation data sheet.

Monitor

1. Click  **Start**, then select **S**ettings, then **C**ontrol Panel
The Control Panel window opens.
2. Double-click **D**isplay to open the Display window.
3. Click the **S**ettings tab and verify that the following configurations are correct:
 - Color depth: **H**ighest
 - Resolution: **1280 x 1024**
 - Font Size: **N**ormal (on the Appearance tab)
 - Refresh Rate: **60 Hz** (On the Settings / Advanced tab)
4. Click **O**K to close the Display window, then close the Control Panel window.
5. Verify that the appearance of the entire monitor display is clean and sharp, and that the color is normal.
6. Check the **Monitor** box in the *Software / Computer* table of the Installation data sheet.

Installing Service Software

To install the required software in the host computer:

1. Insert the **Service CD** i n the host computer drive.
2. In Windows Explorer, navigate to OniSvcToolInstall.exe on the Service CD.
3. Double-click OniSvcToolInstall.exe.

The .exe program installs the files on the host computer. It also installs a program for removing the files, OniSvcToolUninstall.exe.

Note: For detailed information on the service tools installer, see *Installing and Deleting Service Tools*.

Note: When the installation is complete, you must run OniSvcToolUninstall.exe to remove the service software from the host computer. It is policy that all service software must be removed from the host computer before releasing the system to a customer. Leaving the service files on the host computer allows unauthorized changes to system parameters, and could have serious consequences. The service software will be automatically removed 24 hours after it was installed.

RF Loopback Testing

Purpose

The loopback test verifies that the imaging electronics in the electronics cabinet are functioning properly. The test exercises the transmit and receive circuits, including the RF power amplifier.

Requirements

There are no special hardware requirements for this test when performed on systems using the standard CPC RF amplifier (P/N 2005-0016), except that either the 123 mm or 180 mm coil must be installed in the bore. For systems using the Herley RF amp (P/N 2005-0019), the test requires additional hardware. See [Identification and Setup for Herley Amp](#), below. The following software components are required for the loopback test, and are installed by the OniSvcToolInstall.exe program:

- CasWin.exe (spectrometer service interface)
- NSPWINNT.dll (required for CasWin)
- RFLB.dll (loopback pulse sequence)
- Xrstab.exe (analysis program)

Identification and Setup for Herley Amp

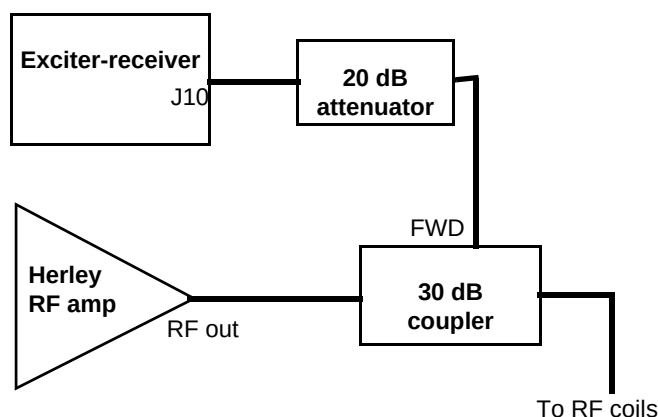
Systems that use a Herley RF amplifier require additional equipment for loopback testing. Refer to the figure below to determine whether the system has a CPC or a Herley amp. A Herley amp has a digital display and diagonal vent slots on the front.

Figure 3-24: Identifying RF Amplifier



If the system has a Herley amp, you must connect a 30 dB coupler and a 20 dB attenuator, as shown at right. If the system has a CPC RF amp, these connections are not necessary.

Figure 3-25: Connections for Testing Herley RF Amplifier



Before running this test, the cabinet must be warmed up by closing the door on the back of the cabinet and powering on all the cabinet components in full-on mode for at least 15 minutes.

Testing RF Loopback

Configuring Loopback Mode and DDS

1. Delete C:\temp\rawdata.dat if it is present on the host computer.
2. Log in to the MR application and enter Service mode.

Figure 3-26: Entering Service Mode

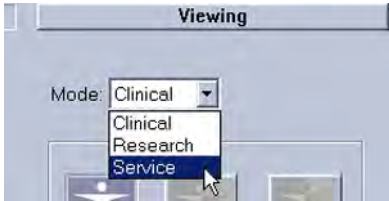


Figure 3-27: Opening CasWin



- a. On the Exam screen, click Service Tools, then click Free CAS in the drop-down menu.
The system frees the CAS bus for use by CasWin.
- b. Click Service Tools, then click CasWin in the drop-down menu.
CasWin launches.
- c. Click Viewing to see the CasWin window.

Figure 3-28: Selecting External Loopback

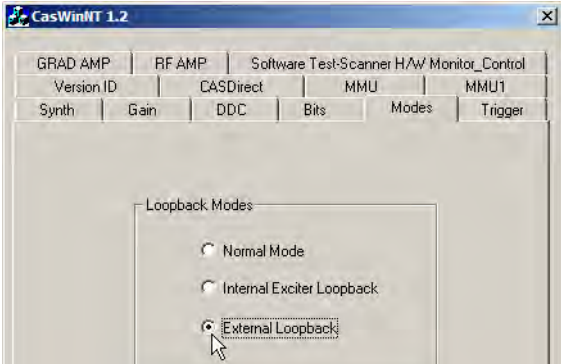
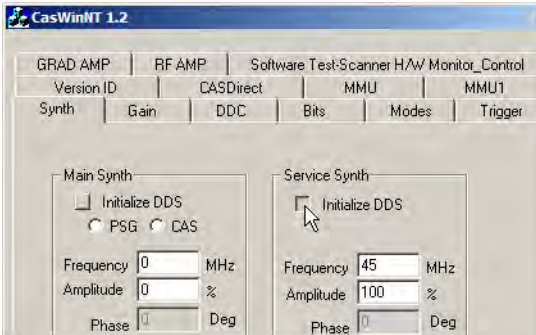


Figure 3-29: Configuring DDS

4. Click the Synth tab, then click Initialize DDS in the Service Synth area.
5. In the Frequency field, enter 45 MHz, and in the Amplitude field, enter 100.
6. Click Quit.
CasWin configures the loopback mode and DDS, then closes.
7. On the Viewing screen, click Exam.
8. Click Service Tools and select Reboot Scan Server from the drop-down menu



Verifying Loopback and Receiver Functions

1. Using the Sequence Editor (click Sequence in the Editor section), create a loopback sequence based on the RFLB pulse sequence. Set the following parameters:

- Freq.: 128
- Phase: 256
- NEX: 8
- TR: 500.

Leave all the other parameters as they are.

Figure 3-30: Selecting Manual Prescan

2. In the Scan Setup window, select the loopback sequence just created, and in the Prescan section, click Manual.

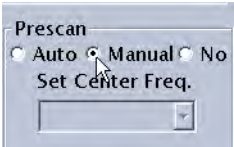


Figure 3-31: Setting Center Frequency

3. Set Center Frequency to 42500000.
4. Under Display Mode X Axis, select Time. (See Figure-3-32.)

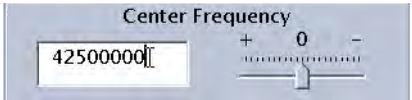
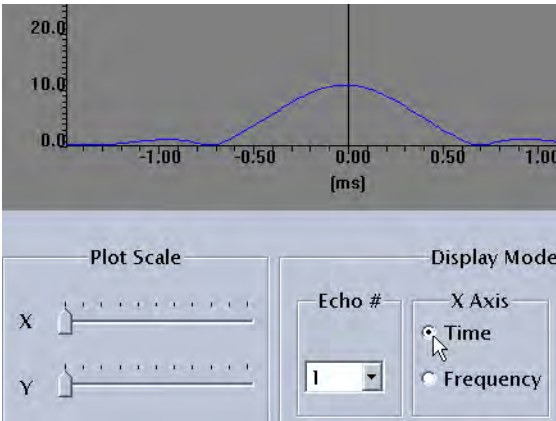


Figure 3-32: Sinc-Shaped Waveform

5. Click Receive Gain and observe the sinc-shaped waveform displayed on the Scan Setup window, at center frequency.
- This verifies that the basic loopback and receiver functions are operating.



Characterizing Stability

1. In the Manual Prescan window, set the Transmit gain to 80.
2. Adjust the Receive gain to a value that produces a signal level that is approximately 50% of Receiver Full Scale.
3. Click Proceed.
4. When the scan is complete, move C:\temp\rawdata.dat to the Service Tools directory. (C:\Program Files\ONI\Orthone\ServiceTools\).
5. From the Service Tools menu, start XRStab.
6. Verify the following field values (update if necessary):
- Raw Data File: rawdata.dat
 - Freq: 128
 - Phase: 2048
 - TR: 500
 - Point No.: 128
 - First N Lines Disregarded: 0
 - Sample Time: 5
7. Click Calculate.

Figure 3-33: RF Loopback Testing Table

8. Record the stability results in the *RF Loopback Testing* table of the Installation data sheet.
9. Exit **XRStab**.

RF Loopback Testing						
Parameter	Units	Min	Max	Measured	P / F	Comments
Signal Mag	%	0	1.000			
Signal Phase	deg	0	1			

Restoring Normal Mode

1. Start **CasWin**.
 - a. Click **Service Tools**, then click **Free CAS**.
The system frees the CAS bus for use by CasWin.
 - b. Click **Service Tools**, then click **CasWin**.
The CasWin window opens.
2. Click the **Modes** tab, and select **Normal Mode**.
3. Click the **Synth** tab, set the **Service Synth Amplitude** to 0, and click **Initialize DDS**.
4. Exit **CasWin**.
5. Select **Reboot Scan Server** from the **Service tools** list.

Gradient Amplifier Calibration

Purpose

This procedure calibrates the gradient amplifier in conjunction with the gradient coil. The calibration includes setting both the sign and amplitude of the configuration parameter for the three linear gradient channels (used for imaging and room temperature shim), and for the B0 channel (used for eddy current compensation). These parameters directly scale the digital gradient waveforms sent to the gradient amplifier. The configuration parameter for each linear gradient channel (X, Y, and Z) is independently adjusted, and corresponds to the DAC value that produces 1.0 Gauss/cm in the imaging system.

Proper completion of the image shim procedure requires gradient calibration.

Hardware Requirements

- A working imaging system
- 123 mm RF coil
- Orientation phantom (Part #2000-0008)

System Polarity

Set the **System_Field_Polarity** parameter in **sysconfig.mdb** to -1.

Image Orientation

1. Insert the 123 mm RF coil.
2. Use the **Localizer** (on the **Manual Prescan** window, click **Localizer**) to center the orientation phantom in the imaging system, with **H** toward the patient end, **A** up, and **L** to the left (when facing the magnet bore from the patient end). The correct position is with the front of the phantom approximately 1.75" (4.4 cm) from the seam between the cylindrical and flared portion of the coil.
3. Set up a study and select **Other** as the body part.
4. Run a sagittal sequence as a scout. (Refer to the *Operator's Guide* for information on sagittal sequences.)
5. Using the sagittal image as reference, run an axial sequence to collect a single slice through the center plastic plate.
6. Verify that the R/L designation in the plate matches the R/L annotation in the image.
If the designations do not match, change the sign of **Grad_Gain_X**.
7. Verify that the A/P designation in the plate matches the A/P annotation in the image.
If the designations do not match, change the sign of **Grad_Gain_Y**.
8. Using the axial image as reference, run a sagittal sequence to collect a single slice through the center plastic plate.
9. Verify that the H/F designation in the plate matches the H/F annotation in the image. If not, change the sign of **Grad_Gain_Z**.
Leave the phantom in position and continue to the next procedure.

B0 Gradient

Set the B0 channel calibration factor to -70,000.

Linear Gradients

Each of the three gradient calibration factors can be set independently. This procedure changes and confirms all three parameters.

1. Set the values of Calibration_Lx, Calibration_Ly, and Calibration_Lz in the system configuration file to match the ID dimensions of the phantom.
2. Restore (using the Protocol Editor) and run the protocol file C:\Program Files\ON\OrthOne\ServiceTools\GradientCal.dat or follow the indented steps below. The protocol file is only available when the Service Tools are loaded.

Note: You may have to slightly offset the phantom along the bore direction to get enough signal for Auto Prescan to properly generate an axial slice.

- a. Set up a SpinEcho sequence with the following parameters:

- Freq.: 256
- Phase: 256
- FOV: 120
- Slice: 3mm
- TR: 300
- TE: min
- BW: 69

- b. From the Manual Prescan window, run Auto Prescan, with Center Frequency method set to Peak, then run Gradient Shim.

- c. Collect an axial image with Freq. Dir set to A/P.

This is referred to below as image #1.

- d. Collect an axial image with Freq. Dir set to R/L.

This is referred to below as image #2.

- e. Collect a sagittal image with Freq. Dir set to A/P.

This is referred to below as image #3.

3. Measure the diameter of image #1 in the R/L direction using the Silhouette line tool. Calculate the new value of Grad_Gain_X, using the following formula:

New value = current value $\times$ (actual ID $\div$ measured ID)

The actual ID is 89 mm.

4. Measure the diameter of image #2 in the A/P direction using the Silhouette line tool. Calculate the new value of Grad_Gain_Y using the formula given in [step 3](#).

5. Measure the diameter of image #3 in the H/F direction using the Silhouette line tool. Calculate the new value of Grad_Gain_Z using the formula given in [step 3](#).

6. If any of the newly calculated parameters differ by more than 1% from current values in the configuration file, update the values in the configuration file and repeat the process.

Only images related to parameters that have been changed must be retaken.

Shim (First Pass)

Refer to the *Magnet Shimming Guide* (DOC0922226) for detailed information on how to perform a shim operation.

Shim the magnet until the peak-to-peak homogeneity of the 12cm $\times$ 15cm volume is less than or equal to 30 ppm.

This is required to increase the magnetic field homogeneity enough to collect good eddy current correction data at locations 4 cm from the magnet center.

Eddy Current Compensation

1. Perform the procedures described in the *Eddy Current Compensation Guide* (DOC0922213), taking the following notes into account.

These procedures characterize the eddy currents produced by the system and change the system configuration files to alter the gradient waveforms, so eddy currents are minimized.

Note: Due to manufacturing variations, a negative B0_Gain calibration value may be incorrect. The following steps check for this condition and correct for it.

Figure 3-34: Gradient Amp Calibration Table

- a. Enter the value specified in **B0 Gradient** on page 62 in the Measured column of the *Gradient Amp Calibration* table of the Installation data sheet for the first iteration of the first major component.
- b. Verify that the B0 component is compensated correctly (Pass appears in the P/F column of the table).
- c. If the B0 component increases after compensation, change the sign of the B0_Gain value.
- d. Restart the eddy analysis tool before continuing, to register the change in the sign of the B0_Gain.

Gradient Amp Calibration						
Gradient Coil Serial #	Parameter	Units	Min	Max	Measured	P / F
	B0 gain	*	*	*		*
	Grad_Gain_X	*	[17,167]	[18973]		
	Grad_Gain_Y	*	[17,167]	[18973]		
	Grad_Gain_Z	*	A: [17,140] B: [18,383]	A: [19,717] B: [20,317]		

2. Include the eddy current correction procedure data sheet with the Installation data sheet.
3. Perform the procedure **Linear Gradients** on page 62 to check the calibration of the linear gradients.

You must recheck the values because changing the eddy current correction can change the linear gradient values. Record the final values of the calibration factors in the *Gradient Amp Calibration* table of the Installation data sheet.

Shim (Second Pass)

1. **Shim** the magnet until the peak-to-peak homogeneity of the 12 cm × 15 cm volume of the shim ellipsoid is less than 15 ppm and the peak-to-peak homogeneity over the 8 × 10 cm volume is less than 2.5 ppm.

This is the final pass, creating a homogenous magnetic field volume. Target shim values listed above are less than the specified values listed in the data sheet, to allow for variations in magnetic field due to room temperature. The larger volume reflects the imaging field of view (FOV), and the smaller volume reflects the FOV for selective fat suppression.

2. Set DiaXY to 150 and DiaZ to 120 in the Shim Ellipsoid field of the Shimming Data tab window.

Figure 3-35: Inhomogeneity Table

3. Click **Calculate**, then read the Error value from the Inhomogeneity over shim ellipsoid table in the Shimming Data tab window. Record the value in the Installation data sheet *Shim and Park Values Table*.

Inhomogeneity over shim ellipsoid		
DiaXY (mm)	DiaZ (mm)	Error (ppm)
150	120	24.38

Figure 3-36: Shim and Park Values Table

4. Read the Error value of the entry DiaXY=150, DiaZ=120 from the Inhomogeneity over ref. ellipsoids table of the Shimming Data tab window. Record the value in the Installation data sheet *Shim and Park Values Table*.
5. Record the Error value of the entry DiaXY=99.1, DiaZ=80 from the Inhomogeneity over ref. ellipsoids table of the Shimming Data tab window. Record the value in the Installation data sheet *Shim and Park Values Table*.

Shim and Park Values						
Parameter	Units	Target	Max	Measured	P / F	Comments
Magnet Park Frequency (before shimming)	MHz	*	*		*	
Magnet Ramp Current (at park frequency)	Amps	*	*		*	
DiaXY=150, DiaZ=120 (shim ellipsoid setting & val)	ppm	15	20			
DiaXY=150, DiaZ=120 (ref. ellipsoid value)	ppm	*	*		*	
DiaXY=100, DiaZ=80 (shim ellipsoid setting & val)	ppm	*	*		*	
DiaXY=99.1, DiaZ=80 (ref. ellipsoid value)	ppm	2.5	3			

Figure 3-37: DiaXY and DiaZ Fields

- Set DiaXY to 100 and DiaZ to 80 in the Shim Ellipsoid fields of the Shimming Data tab window, click Calculate, and read the Error value listed in the Inhomogeneity over shim ellipsoid table of the Shimming Data tab window. Record the value in the Installation data sheet *Shim and Park Values Table*.
- Back up the placement history file for future use. Submit a copy of the file with the Installation data sheet. Leave another copy on a floppy disk at the site.

Shim Ellipsoid (mm)

Dia.XY: 100 Dia.Z: 80

of plot angles: 24

Characterizing the Center Frequency

Figure 3-38: Center Frequency Table

- Install the DQA phantom (P/N 2000-0006) and run auto prescan including gradient shim. (Refer to the *Operator's Guide* for information on prescans.)
- Record the magnet center frequency in the *Center Frequency* table of the Installation data sheet.
- Record the linear shim values returned by the gradient shim algorithm in the *Center Frequency* table of the Installation data sheet.

Center Frequency (after shimming) & Gradient Shim Values						
Parameter	Units	Min	Max	Measured	P / F	Comments
ξ_0	MHz	42.5750	42.6250			
X shim	Hz/cm	-20	20			
Y shim	Hz/cm	-20	20			
Z shim	Hz/cm	-20	20			

Magnet Monitoring Unit

Purpose

This procedure verifies that critical magnet characteristics are properly monitored and records the results. The CasWin tool measures the values being read by the GE MR software. These values are then compared to the values measured directly using a digital multimeter (DMM).

The MMU (P/N 2001-0017) in the GE MR model AB 2000 performs the remote monitoring and alarm functions of the sensaphone in the AA 2000 model and monitors the compressor helium pressure, in addition to monitoring magnet conditions. A procedure for configuring the remote monitoring and alarm functions of the AB 2000 MMU is included in this section ([Configuring MMU Remote Functions](#) on page 66).

Requirements

- An operational system
- CasWin.exe
- NSPWINNT.dll (required for CasWin)

Testing Magnet Monitoring Functions

The magnet temperature measurements are performed using the CasWin service software tool. The tool was installed on the host computer when you performed [Installing Service Software](#) on page 57.

- Log in to the MR application and enter **Service** mode.
- Click **Service Tools**, then click **Free CAS**.

The system frees the CAS bus for use by CasWin.

- Click **Service Tools**, then click **CasWin**.

The CasWin window opens.

- Click the MMU1 tab.

The CasWin MMU1 window opens.

Figure 3-39: Updating MMU1 Window Values

5. Click the Read All button.
- CasWin reads the monitored sensors and conditions, and displays them in the MMU1 window.

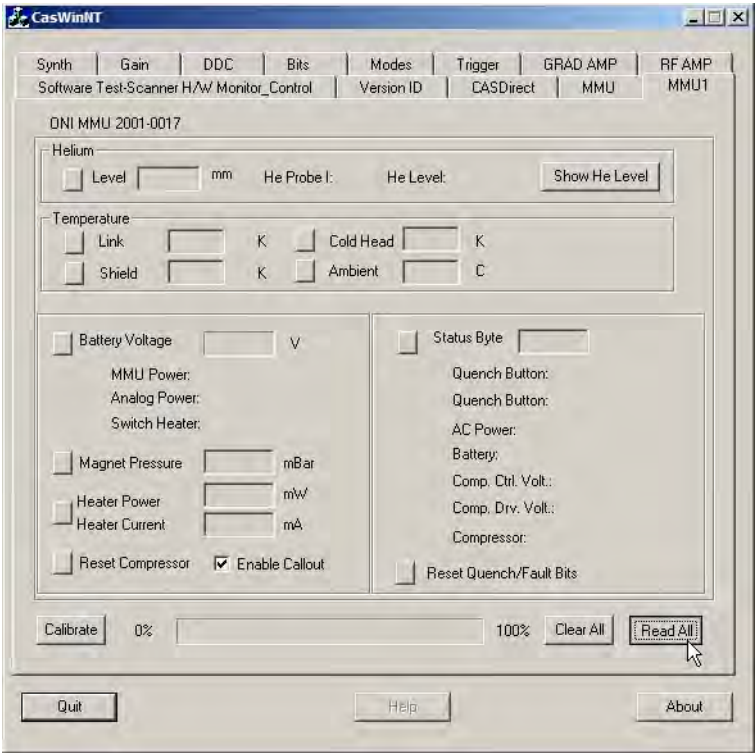


Figure 3-40: Magnet Helium Level and Temperatures Table

6. Record the helium level in the *Magnet Helium Level and Temperatures* table of the Installation data sheet.

Magnet Helium Level and Temperatures						
Function	Units	Min	Max	Measured	P / F	Comments
Helium Level	mm	*	*		*	
Link Temp	K	*	*		*	
Shield Temp	K	*	*		*	
Cold Head Temp	K	*	*		*	

7. Verify that the helium level probe status (He Probe I:) is OK.
8. Record the Link Temperature in the *Magnet Helium Level and Temperatures* table of the Installation data sheet.
9. Record the Shield Temperature in the *Magnet Helium Level and Temperatures* table of the Installation data sheet.
10. Record the Cold Head Temperature in the *Magnet Helium Level and Temperatures* table of the Installation data sheet.
11. Verify that the Battery Voltage value is acceptable.

The minimum acceptable voltage is 18 V.
12. Exit CasWin.
13. Select Reboot Scan Server from the Service Tools list.
14. Verify that the Service Log registers time-stamped entries for shield temperature, link temperature and helium level over five continuous hours.

The log should show readings of these quantities, taken every hour on the hour.

Configuring MMU Remote Functions

This procedure configures the remote monitoring and alarm functions of the AB 2000 model MMU (P/N 2001-0017).

Opening the Dialog Window

1. On the CasWin MMU1 window, click **Calibrate**.

The Dialog window opens. The upper section of the window contains fields for the values that trigger alarm calls. The lower Static section of the window contains fields for the phone numbers the system calls when an alarm is triggered.

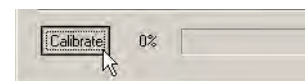


Figure 3-41: Dialog Window

2. Click **Read All**.

Caswin displays the factory default values in the fields associated with the Read button.

3. Compare the screen to [Figure 3-41](#) to make sure all values are set correctly, except for the parameters listed in Table 3-1.

Table 3-1 Customized parameter values

Variable	Threshold High	Threshold Low
Helium Level		Set to ASV - 50
Link	Set to ASV + 10	Set to ASV - 5
Shield	Set to ASV + 10	Set to ASV - 5
Cold Head	Set to ASV + 10	Set to ASV - 5

ASV = The Actual System Value you recorded on the PM data sheet

4. In the Static section of the window, click the  at the right of the **Country Code** field, and select the correct code from the drop-down menu. (See Table 3-2, below.)

This is the modem code for the customer site.

Table 3-2 Modem Country Codes

Country	Code	Country	Code	Country	Code
Argentina	07	Europe	FD	Mexico	B5
Australia	09	Hong Kong	99	New Zealand	7E
Canada	B5	Indonesia	99	Philippines	B5
China	B5	Japan	00	Taiwan	FE
Cyprus	FD	Korea	B5	United States	B5

5. Click **Write**.

CasWin saves the modem code in the configuration file.

6. Enter the **Site ID** (the modem's phone number), and click **Write**. Do not include the 9 that the site may require for dialing out, and do not include the 1 that precedes long-distance calls.

CasWin saves the site ID in the configuration file.

7. Enter each of the phone numbers the unit will call when an alarm trips. Click **Write** after each entry.

CasWin saves the phone numbers in the configuration file.

Figure 3-42: Monitor / Auto Call-Out Device Phone Numbers Table

8. Record the phone numbers, the name associated with each number, and any required country codes in the *Monitor / Auto Call-Out Device Phone Numbers* table of the installation data sheet.
9. On the CasWin Dialog window, click **Test** next to your cell phone number, to verify that the MMU calls out correctly.
10. End the call from the MMU, and call it back, using the site ID phone number, to verify that you can call in to the MMU.
11. On the CasWin Dialog window, click **Done**.
The Dialog window closes.

Monitor / Auto Call-Out Device Phone Numbers				
Item	Contact Name	Dial-out numbers (with all appropriate codes & digits)	Verified	Comments
Dial Out Phone Number 1			☐	
Dial Out Phone Number 2			☐	
Dial Out Phone Number 3			☐	
Dial Out Phone Number 4			☐	

Magnet Pressure Regulator

Purpose

This procedure verifies proper operation of the magnet pressure regulator function of the 2001-0017 MMU.

Preparation

Allow the system to come to thermal equilibrium. This means that the following conditions are met:

- The magnet is ramped
- The compressor has not been turned off or disabled for at least 12 hours

Procedure

Caution: Before performing the following procedure, verify that there is positive pressure in the magnet. Performing the procedure when there is negative pressure in the magnet can cause ice to form in the magnet, leading to problems. If there is negative pressure, turn off the compressor for 15 minutes, then restart the compressor and proceed.

1. On the CasWin MMU1 window, note the Magnet Pressure value. (See [Figure-3-39](#) on page 65.)

Figure 3-43: Magnet Pressure Regulator (helium port closed) Table

2. Record the Magnet Pressure value in the *Magnet Pressure Regulator (helium port closed)* table of the data sheet.
3. Remove the helium fill port cap and place a tissue in the open port to prevent air from entering the magnet.

This brings the pressure inside the magnet to atmosphere and allows verification of proper calibration of the pressure regulator.

4. On the CasWin MMU1 window, click **Update All**.
5. Verify that the Magnet Pressure value is less than the value noted in [step 1](#), and that the value is between -4 and +4 mBar.
6. Verify that the Heater Current is between 87 mA and 93 mA.
7. If the Heater Current value is not between 87 mA and 93 mA, adjust the heater current pot on the MMU circuit board.
 - a. Shut off the power switch at the rear of the MMU.
 - b. Disconnect the MMU batteries, using the slide switch at the front of the MMU.
 - c. Disconnect all the cables from the rear of the MMU. (Note their positions.)
 - d. Remove the four screws that secure the front of the MMU to the electronics cabinet.
 - e. Place a towel or a piece of cardboard on top of the electronics cabinet, to protect the paint.
 - f. Slide the MMU out the front of the electronics cabinet, and place the MMU on top of the cabinet.
 - g. Reconnect all the cables to the rear of the MMU.
 - h. Turn on the power switch at the rear of the MMU.
 - i. Remove the top panel of the MMU.

Magnet Pressure Regulator (helium fill port closed)						
Item	Units	Min.	Max.	Measured	P / F	Comments
Magnet Pressure	mbar	7.50	74.00			
Heater Power	mW	300	1000			
Heater Current	mA	*	*		*	

- j. Adjust the VR2 pot on the MMU main circuit board slightly.
- k. On the CasWin MMU1 window, click **Magnet Pressure**.
The Heater Current reading updates.
- l. Repeat [step j](#) and [step k](#) until the Heater Current value displayed on the CasWin MMU1 window is between 87 mA and 93 mA.
- m. Reverse [step i](#) through [step a](#) to restore the MMU to operating condition.
- n. Verify that the switch at the front of the MMU is set so the MMU batteries are connected.

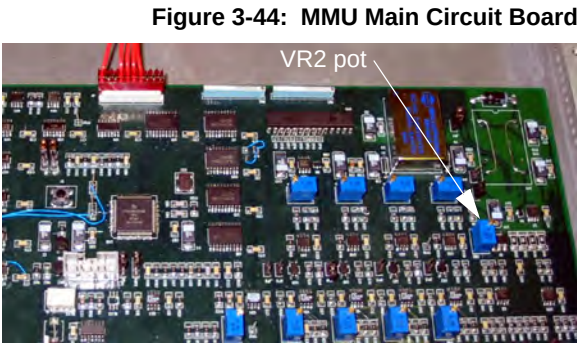


Figure 3-45: Magnet Pressure Regulator (helium port open) Table

- 8. Record the Magnet Pressure value in the *Magnet Pressure Regulator (helium port open)* table of the data sheet.
- 9. Replace the helium port cap.
- 10. On the MMU1 Calibrate window, set the Magnet Pressure Low value to **5 mBar higher** than the value obtained in [step 8](#), and the Magnet Pressure High value to **69 mBar higher** than the value obtained in [step 8](#).

Magnet Pressure Regulator (helium fill port open)						
Item	Units	Min.	Max.	Measured	P / F	Comments
Magnet Pressure	mbar	-4	4			
Heater Current	mA	87	93			

Characterizing the Field Stability

Purpose

This procedure characterizes the stability of the magnetic field, using point sample measurements. The change in magnetic field as a function of frequency is plotted out by processing the collected short TE/short TR gradient echo data. Large changes in the magnetic field can cause ghosting in the images. Changes in magnet field can arise from site instability, improper dampening of the cold-head motion, improperly placed magnet covers, improperly routed helium coldhead lines, or stray signals from the electronics cabinet. Several positions are measured, because the instability may be a function of position in the magnet.

Requirements

- A functional imaging system
- Point sample apparatus
- Coil ID jumper
- 50 Ω terminator
- Variable attenuator
- BNC coax cable
- GE2D_pete.dll file
- Magnet Stability.exe, in, and frequency plot.xls files

Note: The magnet covers must be installed before performing this procedure.

Procedure

The GE2D_pete.dll file is used in this procedure.

- 1. Place the variable attenuator at RF out on the back of the Exciter/Receiver.
- 2. Replace the Coil ID cable at the RFFE with the Coil ID jumper.
- 3. Disconnect the RF cables at the RFFE. Attach the 50 Ω terminator to one output on the front end. Attach the point sample cable to the other output.
- 4. Start the MR Scanner application, and enter **Service** mode.
- 5. Edit the sysconfig.mdb file, and set the four Grad_Gain parameters in the SystemConfig table to 0.
- 6. Use the protocol editor to load the SmallSample sequence.

Note: The SmallSample sequence is based on the GE2D_pete pulse sequence, using the following parameters:

- Freq.: 8
- Phase: 128
- FOV: 160
- NEX: 64
- TR/TE: 6.1/3.1
- FA: 30
- Slices: 1
- BW: 15
- Prescan: No

7. Position the sample apparatus in the gradient coil.
8. Set the attenuation at the back of the Exciter/Receiver, the Receiver Gain, and the Transmit Gain to values that maximize the signal-to-noise ratio of the GE2D_pete sequence.
Typical values are: attenuation = 40 dB, Receiver Gain = 100, Transmit Gain = 100.
9. Run the stability sequence with the sample at all of the positions listed below. Rename the raw data file to reflect the sequence used and the sample location.
 - (0, 0, 0) Gradient isocenter
 - (4, 0, 0) +4cm along the X (R/L) axis from isocenter (right)
 - (-4, 0, 0) -4cm along the X (R/L) axis from isocenter (left)
 - (0, 4, 0) +4cm along the Y (A/P) axis from isocenter (up)
 - (0, -4, 0) -4cm along the Y (A/P) axis from isocenter (down)
 - (0, 0, 4) +4cm along the Z (H/F) axis from isocenter (patient)
 - (0, 0, -4) -4cm along the Z (H/F) axis from isocenter (service)
10. Work up and plot the data collected.
Refer to the *Stability Workup Procedure* for instructions.
11. Submit the plots with the completed Installation data sheet.
12. Remove the point sample apparatus and reattach the RF coil cables and the Coil ID cable.
13. Remove the variable attenuator.
14. Reset the Grad_Gain values in the sysconfig.mdb file to the calibrated values.

Characterizing the 180 mm RF Coil

This procedure characterizes the Signal-to-Noise ratio (SNR) of the 180 mm RF coil. The procedure is also a system test that verifies the calibration of the transmit and receive chains. The Daily QA (DQA) boundary values are set based on the results of this test. These values are then used in any DQA test as additional pass/fail criteria, to indicate if a significant change has occurred in the system causing a drastic (>10%) difference in the prescan parameters.

1. Verify that the Daily QA SNR pass/fail criteria in the system configuration file are set to the values listed in the data sheet for the 180 mm RF coil.
2. Insert the 180 mm RF coil into the magnet bore.
3. Place the DQA phantom in the magnet bore.
4. Select the Daily QA protocol, set the sequence to manual prescan, and click **Begin Protocol**.
5. In the Manual Prescan window, run auto prescan with the Center Freq option set to **peak**, then run Gradient Shim.

Figure 3-46: 180 mm RF Coil Table

6. Record the Transmit gain and Receive gain values obtained by the prescan algorithm in the *180 mm RF Coil* table of the Installation data sheet.
7. Record the % of Receiver Full Scale value displayed in the field when in Receive Gain mode in the *180 mm RF Coil* table of the Installation data sheet.
8. Click **Proceed** to start data acquisition.
9. When the scan is complete, record the following values in the *180 mm RF Coil* table of the Installation data sheet:
 - Sig Mean
 - Noise STD
 - Noise Mean
 - Ghost Mean
 - ASR

Parameter	Units	Min	Max	Measured	P / F	Comments
TG	1/10 db steps	120	160			
RG	1/10 db steps	*	*		*	
%Recv	%	30	52			
Sig Mean	*	*	*		*	
Noise STD	*	*	*		*	
Noise Mean	*	*	*		*	
Ghost Mean	*	*	*		*	
SNR	*	110	140			
ASR	%	0	1			

The data sheet automatically calculates the signal-to-noise ratio, and indicates whether it is within specifications.

Figure 3-47: Site Information Table

10. Multiply the RG value set in [step 5](#) by 0.9 to calculate the `Daily_QA_RX_Gain_Lower` value. Round down to the nearest integer, and record the value in the *Site Information* table of the Installation data sheet.
11. Multiply the RG value set in [step 5](#) by 1.1 to calculate the `Daily_QA_RX_Gain_Upper` value. Round up to the nearest integer. If the result is greater than 200, set the value to 200. Record the value in the *Site Information* table of the Installation data sheet.
12. Multiply the TG value set in [step 5](#) by 0.9 to calculate the `Daily_QA_TX_Gain_Lower` value. Round down to the nearest integer, and record the value in the *Site Information* table of the Installation data sheet.
13. Multiply the TG value set in [step 5](#) by 1.1 to calculate the `Daily_QA_TX_Gain_Upper` value. Round up to the nearest integer. If the result is greater than 200, set the value to 200. Record the value in the *Site Information* table of the Installation data sheet.
14. Update `sysconfig.mdb` with the values calculated in [step 10](#) through [step 13](#).

Item	Value	Comments
Daily_QA_RX_Gain_Lower		
Daily_QA_RX_Gain_Upper		
Daily_QA_TX_Gain_Lower		
Daily_QA_TX_Gain_Upper		
RfAmp_Trip_Pt_Slope		
Institution Name		
Institution Address		
RX_DAC_Slope		
RX_DAC_Start		
Scanner_Serial_Num		
Station		
TX_DAC_Slope		
TX_DAC_Start		
UI_Date_Format		

Note: To make changes to the `sysconfig.mdb` file take effect, you must log off or reboot the host computer.

Characterizing the 123 mm RF Coil

This procedure characterizes the Signal-to-Noise ratio (SNR) of the 123 mm RF coil. This procedure is also a system test that verifies the calibration of the transmit and receive chains.

1. Insert the 123 mm RF coil into the system.
2. Place the DQA phantom in the magnet bore.
3. Select the **Daily QA** protocol.
4. Set the following parameters:
 - Slice thickness: 3.5mm
 - Freq: 512
 - Bandwidth: 50
5. Set the sequence to manual prescan, and click **Begin Protocol**.

- In the Manual Prescan window, run auto prescan, with the Center Freq option set to **peak**, then run Gradient Shim.

Figure 3-48: 123 mm RF Coil Table

- Record the values for Transmit gain and Receive gain obtained by the prescan algorithm in the *123 mm RF Coil* table of the Installation data sheet.
- Record the %Receiver value displayed in the field when in the Receive Gain mode in the *123 mm RF Coil* table of the Installation data sheet.
- Click **Proceed** to start data acquisition.
- When the scan is complete, record the following values in the *123 mm RF Coil* table of the Installation data sheet:
 - Sig Mean
 - Noise STD
 - Noise Mean
 - Ghost Mean
 - ASR

123mm RF Coil						
Parameter	Units	Min	Max	Measured	P/F	Comments
TG	1/10 db steps	40	60			
RG	1/10 db steps	*	*		*	
%Recv	%	*	*		*	
Sig Mean	*	*	*		*	
Noise STD	*	*	*		*	
Noise Mean	*	*	*		*	
Ghost Mean	*	*	*		*	
SNR	*	114	136			
ASR	%	0	1			

The data sheet automatically calculates the signal-to-noise ratio, and indicates whether it is within specifications.

Characterizing Optional RF Coils

This procedure characterizes the signal-to-noise ratio (SNR) of an RF coil. The procedure is also a system test that verifies the calibration of the transmit and receive chains. The Daily QA (DQA) boundary values are listed in the coil's table in the Installation data sheet. The prescan values can be used in a DQA test as additional pass/fail criteria, to indicate if a significant change has occurred in the system causing a drastic (>10%) difference in the prescan parameters.

- Insert the RF coil into the magnet bore.
- Place the DQA phantom in the magnet bore.

Note: Coils with a diameter less than 100 mm use a smaller phantom. Each coil uses its own sleeve to position the phantom inside the coil.

- For coils with a diameter of 160 mm or greater, select the Daily QA protocol.
For coils with a diameter less than 160 mm, select the Daily QA 123 protocol.
- Set the sequence to manual prescan, and click **Begin Protocol**.
- In the Manual Prescan window, run auto prescan with the Center Freq option set to **peak**, then run Gradient Shim.
- Record the Transmit gain and Receive gain values obtained by the prescan algorithm in the appropriate table of the Installation data sheet.
- Record the % of Receiver Full Scale value displayed when in Receive Gain mode.
- Click **Proceed** to start data acquisition.
- When the scan is complete, record the following values in the Installation data sheet:
 - Sig Mean
 - Noise STD

The data sheet automatically calculates the signal-to-noise ratio, and indicates whether it is within specifications.

Characterizing the Helium Compressor

1. Verify that the compressor and coldhead power cables are installed with proper strain reliefs.

Figure 3-49: Helium Compressor Measurements Table

Helium Compressor Measurements						
Parameter	Units	Min	Max	Measured	P / F	Comments
Pressure (static)	MPa	1.60	1.65			
Pressure (dynamic)	MPa	2.00	2.30			
Hours	*	*	*		*	

2. Read the pressure gauge on the back of the compressor when the compressor is off, and record the value in the *Helium Compressor Measurements* table of the Installation data sheet. (See Figure-2-40 on page 29 for the location of the pressure gauge.)

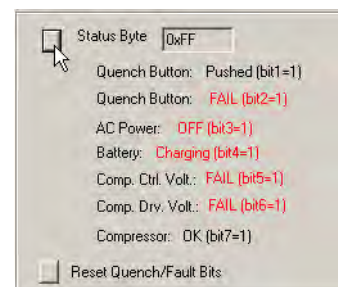
This is the static pressure.

3. Read the pressure gauge on the back of the compressor when the compressor is on, and record the value in the *Helium Compressor Measurements* table of the Installation data sheet.

This is the dynamic pressure.

4. Record the compressor hours displayed on the hours meter (see Figure-2-40 on page 29) in the *Helium Compressor Measurements* table of the Installation data sheet.

Figure 3-50: CasWin MMU1 Byte Status



5. On the CasWin MMU1 window, click **Status Byte**.

CasWin reads several conditions and displays their status.

Note: In the figure, most of the conditions are shown failing, in red. When the conditions are passing, the values display in black text.

Figure 3-51: Helium Compressor Status Table

Helium Compressor Status		
Function	Check	Comments
Compressor Control Voltage OK	☐	
Compressor Drive Voltage OK	☐	
Compressor OK	☐	

6. Record the values for Comp. Ctrl. Volt.: and Comp. Drv. Volt.:, and the Compressor: status (OK or Fail) in the *Helium Compressor Status* table.

Miscellaneous Tests

Laser Alignment Light

Figure 3-52: Miscellaneous Checks table

Miscellaneous Checks		
Function	Check	Comments
System Labeling	☐	
MRI Warning Signs	☐	
Laser Alignment Light	☐	
Patient Chair	☐	
Magnet Covers	☐	
Coil ID	☐	
Room RFI	☐	
Power Distribution Unit (PDU)	☐	
Quench Button	☐	
Safety Grounds	☐	
Equipment room temp	☐	
Magnet room temp	☐	
Configuration Backup	☐	

1. Verify that the laser light is attached securely to the magnet.
2. Verify that the laser light switch turns the laser on and off properly.
3. Verify that if the light is left on, initiating a scan turns the light off.
4. Aim the laser so that the laser line projects properly.

The distance of the laser plane from isocenter should be 9.8 inches (248 mm), and the line should be parallel with the front face of the magnet covers.

5. Check the **Laser Alignment Light** box in the *Miscellaneous Checks* table of the Installation data sheet.

Patient Chair

The *Preventive Maintenance Guide* includes detailed checks of the chair functions.

1. Verify that the following items function as designed:
 - Seat back recline lever
 - Seat recline lever
 - Foot rest
 - Chair brake
 - Chair wheels
2. Verify that there are no physical defects.
3. Verify that there is little or no play in the chair seat or seat back.
4. Check the **Patient Chair** box in the *Miscellaneous Checks table* of the Installation data sheet.

Magnet Covers

1. Verify that the magnet covers fit properly in relation to the magnet, its individual pieces, the gradient coil, and the RF coils.
2. Verify that there are no physical defects.
3. Check the **Magnet Covers** box in the *Miscellaneous Checks table* of the Installation data sheet.

Coil ID

1. Verify that the GE MR software recognizes the 180 mm RF coil and properly annotates the images. (Refer to the *Operator's Guide* for information on image annotations.)
2. Verify that the GE MR software recognizes the 123 mm RF coil and properly annotates the images.
3. Verify that when a scan is started without a coil properly inserted into the coil connector, an error message displays.
4. Check the **Coil ID** box in the *Miscellaneous Checks table* of the Installation data sheet.

RF Interference

1. Install the 123 mm RF Coil. Remove any phantoms from the coil.
2. Start the system software in **Service Mode** and select the **RFI sequence**.
3. Turn off image masking by setting the Valid_Space_Rmax and Valid_Space_Zmax system configuration variables to 0.2.
4. Set the following scan parameter values. (Write down the existing settings; you will reset them when the test is done.)
 - Freq.: 256
 - Phase: 256
 - FOV: 160
 - BW: 80
 - Echoes: 1
 - TR: 50
 - TE: minimum
 - Slices: 1
 - Thickness: 5
5. Open the **Manual Prescan** window.
6. Set the following configuration values. (Write down the existing settings; you will reset them when the test is done.)
 - Transmit gain: 120
 - Receive gain: 200
 - Center frequency: 42100000
7. Click **Proceed** and verify that there are no perceptible RFI artifacts (zippers, coherent dots, etc.) in the resulting noise image. (See *Optimizing Image Quality* for examples of these artifacts.)
8. Increase the center frequency by 75kHz and repeat the scan, verifying that the image is clean. Repeat the process at 15 frequencies, up to a center frequency of 43.15 MHz.

9. Check the **Room RFI** box in the *Miscellaneous Checks table* of the Installation data sheet.
10. Reset the system configuration variables to the values you wrote down in [step 4](#) and [step 6](#).

Power Distribution Unit

Perform these tests with the power switches of all electronics rack components on, and both circuit breakers on the I/O panel in the On position.

1. Verify that when the system is in **Full On mode**, all components are powered on.
2. Verify that when the system is in **Overnight mode**, only the scanner computer, Digital IO and Exciter/Receiver are powered on.
3. Verify that when the system is in **Emergency Stop** mode, only the scanner computer, Digital IO and Exciter/Receiver are powered on.
4. Check the **Power Distribution (PDU)** box in the *Miscellaneous Checks table* of the Installation data sheet.

Safety Grounds

1. Use a DMM to verify that both **safety ground cables** are installed properly, with no resistance to chassis ground.

The safety ground cables are:

- From the magnet to the penetration panel
- From the penetration panel to the cabinet

2. Check the **Safety Grounds** box in the *Miscellaneous Checks table* of the Installation data sheet.

Equipment Room Temperature

1. Verify that the average temperature in the equipment room is between 18 °C and 25° C, and does not change by more than 3° C during an hour.
2. Check the **Equipment room temp** box in the *Miscellaneous Checks table* of the Installation data sheet.

Magnet Room Temperature

1. Verify that the average temperature in the Exam room is between 18 C and 25° C, and does not change by more than 3° C during an hour.
2. Check the **Magnet room temp** box in the *Miscellaneous Checks table* of the Installation data sheet.

Characterizing Image Quality

1. Center the orientation phantom in the 123 mm RF coil. The approximate location is with the front edge of the phantom 1.75” (4.4 cm) from the seam between the flared part and the cylindrical part of the coil.
2. Use the **Protocol Editor** to load (restore) the protocol and sequences in ImageQuality.dat or go to the manual scan description in step 3.

Note: ImageQuality.dat was copied to the host computer when you ran OniSvcToolInstall.exe. (See [Installing Service Software](#) on page 57.) You cannot restore protocols or sequences directly from the CD; you must restore them from the hard disk.

- a. On the screen, click Exam.
- b. Click **Protocol Editor**.
The Protocol and Sequence Editor window opens.
- c. Click **Backup**.
- d. Click **Browse**, and browse to C:\Program Files\ONI\orthone\Service Tools\ImageQuality.dat.
- e. Click **Open**.
The browser window closes.
- f. On the Exam window, click **Restore**.
The window displays a list of the restored sequences.

g. Click **Done**.

The Protocol and Sequence Editor window closes.

- Run the Image Quality protocol or manually create and run the following six **image quality sequences** to characterize the image ghosting artifacts. Run each sequence once, to obtain nine slices for each sequence (18 slices for the IQ2 sequence).

Sequence Name	Base Sequence	Freq.xPhase	TR	TE	FOV	NEX	BW	ETL	Slices	Thick	Gap	Orient	Freq
IQ1	SE	256x192	350	20	160	1	50	-	9	4	1	Ax	L/R
IQ2	SE	256x192	1500	20/80 ^a	160	1	50	-	9	4	1	Ax	L/R
1Q3	FSE	256x192	1500	(min)	160	1	50	8	9	4	1	Ax	L/R
1Q4	FSE	256x192	1500	(min)	160	1	50	8	9	4	1	Sag	A/P
IQ5	FSE	256x192	1500	(min)	160	1	50	8	9	4	1	Cor	L/R
IQ6	FSE	256x192	1500	80	160	1	50	8	9	4	1	Ax	L/R

a. The IQ2 sequence acquires two echoes, and generates two nine-slice series.

Figure 3-53: Image Quality Table

- Calculate the ghosting percentage for each slice by dividing the difference of the mean of the ghost intensity and the noise intensity by the average of the signal intensity.

$$100 \times (\text{GhostMean} - \text{NoiseMean}) \div \text{SigMean}$$

- Record the highest slice value for each sequence in the *Image Quality* table of the Installation data sheet.

Image Quality				
Pulse Sequence	Units	Orientation	% ghosting	Comments
T1-weighted SE (TE=20)	%	Axial		
Dual Echo SE (TE=20/80)	%	Axial		
PD-weighted FSE	%	Axial		
PD-weighted FSE	%	Sagittal		
PD-weighted FSE	%	Coronal		
T2-weighted FSE	%	Axial		

Modem Dial-In

- With the scanner PC powered on, have someone log on remotely to the scanner.
- Have them verify that files can be copied from the scanner system and placed onto the scanner system.
- Record the telephone number used to connect to the modem in the *Network Configuration* table of the Installation data sheet.

Figure 3-54: Network Configuration Table

Network Configuration		
Item	Value	Comments
System Phone Number		
Host Computer's Network Name		
Host Computer's IP Address		
Domain Name (if any)		
Subnet Mask Value		
Gateway Address (if any)		

Figure 3-55: Software / Computer Table

- Check the **Modem dialin** box in the *Software / Computer* table of the Installation data sheet.

Software / Computer		
Software Version #		
Function	Check	Comments
ONI SW install	☐	
Modem dial-in test	☐	
Image Archive test	☐	
Monitor check	☐	

Image Archive

- Update the DVD drive letter in the SCSI.fol file to match the drive letter of the MaxOptix drive listed in Windows Explorer.
- Collect a series of images and verify that they can be archived to a CD or a DVD.
- Record satisfactory completion of Image Archive in the *Software / Computer* table of the Installation data sheet.

DICOM Transfer

- Update the following parameters in the DCNodes file (refer to Configuring the DCnodes File in the *Silhouette Service Manual* for instructions):
 - Host: [The host name for the DICOM device] (from the site network administrator)
 - Port: 4444
 - MaxLength: 65536
 - Alias: [A descriptive name for the DICOM device]
- Verify that images can be transferred successfully to all applicable remote sites, using DICOM transfer.
 - Use the Windows Task Manager to check that the DICOM server **dcserver** is running.
See Displaying Active Processes in the *Silhouette Service Manual* for details on how to view running processes.
 - Initiate an image transfer from a remote workstation or scanner.

Note: DICOM sends the images directly to the workstation database

- Open the patient folders on the host computer, and check that the images are not corrupted.

Figure 3-56: DICOM Transfer Configuration & Test Table

DICOM Transfer Configuration & Test (Parameter data taken from DCNodes file, must match exactly)				
Parameter	Site #1	Site #2	Site #3	Comments
AE Title				
IP Address				
Port Number				
MaxLength (65536 default)				
Alias (if any)				

- Record the values of the parameters updated in [step 1](#) in the *DICOM Transfer Configuration & Test* table of the Installation data sheet.
- Check the **DICOM Transfer** box in the *Software / Computer* table of the Installation data sheet.

Configuration Backup

Figure 3-57: Miscellaneous Checks table

Miscellaneous Checks		
Function	Check	Comments
System Labeling	☐	
MRI Warning Signs	☐	
Laser Alignment Light	☐	
Patient Chair	☐	
Magnet Covers	☐	
Coil ID	☐	
Room RFI	☐	
Power Distribution Unit (PDU)	☐	
Quench Button	☐	
Safety Grounds	☐	
Equipment room temp	☐	
Magnet room temp	☐	
Configuration Backup	☐	

- When testing is complete and all systems meet specification, run the **BackUpConfig** utility to collect all the configuration files.
See Backing Up the System Configuration Files in the *Scanner Software v.2 Installation Guide* for instructions on running the BackUpConfig utility.
- Copy the configuration files onto a floppy disk. You may also want to make a copy on the D: drive.
- Label the disk with the site name and the date, and leave the floppy disk with the site software.
- Copy the shim history file onto another floppy disk, label the disk, and leave it with the site software.
- Record completion in the *Miscellaneous Checks table* of the Installation data sheet.
- Email copies of the **Installation data sheet**, **System stability .pdf file** and the **Configuration files directory (BackupTemp)** to **SpecialtyMRmsk@ge.com**.

Removing Service Software

When the service is complete, you must run OniSvcToolUninstall.exe to remove the service software from the host computer. It is GE Healthcare policy that all service software must be removed from the host computer before releasing the system to a customer. Leaving the service files on the host computer allows unauthorized changes to system parameters, and could have serious consequences. The service software will be removed automatically 24 hours after it was installed.

To remove the service software:

1. In Service mode, select OniSvcToolUninstall.exe from the Service Tools menu.

The .exe program removes the service files from the host computer.

Note: For detailed information on the uninstall tool, see *Installing and Deleting Service Tools*.

2. Select Reboot Scan Server from the Service tools list.

Appendix A Reference Tables

This appendix contains tables that list the following information:

- [Ten-Pin Sealed Connector Pin Labeling](#) on page 78
- [Service Turret Connector Pinouts](#) on page 78
- [Vacuum Valve Port Pinouts](#) on page 79
- [Cryocooler Turret Connector #1](#) on page 79
- [Helium Depth-to-Volume Conversions](#) on page 79
- [Allen-Bradley Resistance-to-Temperature Conversions](#) on page 80
- [CLTS Resistance-to-Temperature Conversions](#) on page 81

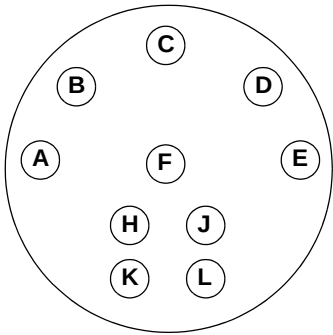
Connector Pinouts

Note: In the tables in this section, the signal notations V+ and V– refer to positive and negative voltages measured between two pins. The notations I+ and I– refer to currents applied to the pin.

Ten-Pin Sealed Connector Pin Labeling

Figure A-1 shows the pin labeling in ten-pin sealed connectors. These connectors include the vacuum valve ports, the service turret connector, and the cryocooler connector.

Figure A-1: Sealed Connector Pinouts



Service Turret Connector Pinouts

Table A-1 lists the pinouts of the service turret connector.

Table A-1Service Turret Connector Pinouts

Pin	Description	Signal
A	Helium level probe 1	V+
B	Helium level probe 1	I+
C	Helium level probe 1	V–
	Helium level probe 2 (backup)	
D	Helium level probe 1	I–
	Helium level probe 2 (backup)	
*	Helium level probe 2 (backup)	V+
*	Helium level probe 2 (backup)	I+
E	Switch heater 1	
F	Common	
H	Allen Bradley sensor	V+
J	Allen Bradley sensor	V–
K	Magnet voltage	V+
L	Magnet voltage	V–

*The backup helium level probe is provided in case the primary level probe fails. The backup probe leads for V+ and I+ are not connected to the connector.

Vacuum Valve Port Pinouts

Table A-2 lists the pinouts of the two vacuum valve ports.

Table A-2 Vacuum Valve Port Pinouts

Port #1			Port #2		
Pin	Description	Signal	Pin	Description	Signal
A	CLTS - link temperature 1	I+	A	CLTS - link temperature 2	I+
B	CLTS - link temperature 1	V+	B	CLTS - link temperature 2	V+
C			C		
D	CLTS - link temperature 1	V-	D	CLTS - link temperature 2	V-
E	CLTS - link temperature 1	I-	E	CLTS - link temperature sensor 2	I-
F			F		
H	CLTS - shield temperature end plate (service end, bottom)	I+	H	CLTS - shield temperature end plate (patient end, top)	I+
J	CLTS - shield temperature end plate (service end, bottom)	V+	J	CLTS - shield temperature end plate (patient end, top)	V+
K	CLTS - shield temperature end plate (service end, bottom)	V-	K	CLTS - shield temperature end plate (patient end, top)	V-
L	CLTS - shield temperature end plate (service end, bottom)	I-	L	CLTS - shield temperature end plate (patient end, top)	I-

Cryocooler Turret Connector #1

Table A-3 lists the pinouts of cryocooler turret connector #1.

Table A-3 Cryocooler Turret Connector #1 Pinouts

Pin	Description	Signal
A		
B		
C	Pressure Control Heater	
D		
E		
F	Pressure Control Heater	
H	CLTS - coldhead 1st stage	I+
J	CLTS - coldhead 1st stage	V+
K	CLTS - coldhead 1st stage	V-
L	CLTS - coldhead 1st stage	I-

Measurement Conversions

Helium Depth-to-Volume Conversions

Table A-4 shows the volume of liquid helium remaining in the magnet at various depth readings.

Table A-4 Helium Depth-to-Volume Conversions

Level Monitor Reading (mm)	Liters Remaining	Level Monitor Reading (mm)	Liters Remaining	Level Monitor Reading (mm)	Liters Remaining
0	1.4	260	8.1	520	16.2
20	1.4	280	8.8	540	16.8
40	1.6	300	9.4	560	17.5
60	1.9	320	10.0	580	18.2
80	2.3	340	10.6	600	18.7
100	3.0	360	11.3	620	19.5
120	3.6	380	11.9	640	20.1
140	4.3	400	12.5	660	20.8
160	5.0	420	13.1	680	21.1
180	5.6	440	13.8	700	21.4
200	6.3	460	14.4	720	21.5
220	6.9	480	15.0		
240	7.5	500	15.6		

Allen-Bradley Resistance-to-Temperature Conversions

The resistance of the Allen-Bradley temperature sensor decreases as its temperature rises. Table A-5 lists the temperatures that correspond to various resistances of the Allen-Bradley temperature sensor.

Table A-5 Allen-Bradley Resistance-to-Temperature Conversions

A-B Sensor Reading (Ω)	Temperature (Kelvin)	A-B Sensor Reading (Ω)	Temperature (Kelvin)	A-B Sensor Reading (Ω)	Temperature (Kelvin)	A-B Sensor Reading (Ω)	Temperature (Kelvin)
20948	2.32	7347	3.30	4079	4.26	407.52	55
20347	2.34	7236	3.32	4041	4.28	404.74	56
19770	2.36	7127	3.34	4003	4.30	402.03	57
19217	2.38	7020	3.36	3966	4.32	399.42	58
18687	2.40	6916	3.38	1106.1	11	396.89	59
18177	2.42	6815	3.40	1016	12	394.44	60
17687	2.44	6716	3.42	944.39	13	392.07	61
17217	2.46	6619	3.44	885.69	14	389.77	62
16765	2.48	6524	3.46	836.79	15	387.54	63
16330	2.50	6432	3.48	795.40	16	385.36	64
15911	2.52	6342	3.50	759.91	17	383.25	65
15508	2.54	6253	3.52	729.14	18	381.21	66
15120	2.56	6167	3.54	702.21	19	379.22	67
14745	2.58	6083	3.56	678.38	20	377.30	68
14386	2.60	6000	3.58	657.16	21	375.41	69
14043	2.62	5919	3.60	638.55	22	373.57	70
13725	2.64	5840	3.62	621.49	23	371.79	71
13122	2.68	5763	3.64	605.79	24	370.05	72
12835	2.70	5689	3.66	591.31	25	368.36	73
12558	2.72	5619	3.68	577.91	26	366.72	74
12289	2.74	5550	3.70	565.83	27	365.10	75
12029	2.76	5483	3.72	554.79	28	363.53	76
11777	2.78	5417	3.74	544.47	29	362.00	77
11533	2.80	5352	3.76	534.80	30	360.51	78
11297	2.82	5289	3.78	525.72	31	359.05	79
11068	2.84	5227	3.80	517.18	32	357.64	80
10846	2.86	5166	3.82	509.33	33	356.24	81
10631	2.88	5106	3.84	501.95	34	354.88	82
10422	2.90	5047	3.86	494.97	35	353.55	83
10219	2.92	4990	3.88	488.35	36	352.25	84
10023	2.94	4933	3.90	482.06	37	350.98	85
9832	2.96	4878	3.92	476.09	38	349.74	86
9646	2.98	4823	3.94	470.48	39	348.53	87
9466	3.00	4770	3.96	465.14	40	347.34	88
9292	3.02	4718	3.98	460.05	41	346.18	89
9122	3.04	4666	4.00	455.18	42	345.04	90
8956	3.06	4616	4.02	450.53	43	343.93	91
8796	3.08	4566	4.04	446.07	44	342.84	92
8640	3.10	4517	4.06	441.83	45	341.77	93
8488	3.12	4469	4.08	437.77	46	340.72	94
8348	3.14	4422	4.10	433.87	47	339.70	95
8212	3.16	4376	4.12	430.12	48	338.70	96
8079	3.18	4331	4.14	426.52	49	337.72	97
7949	3.20	4286	4.16	423.06	50	336.58	98
7823	3.22	4243	4.18	419.72	51	335.40	99
7699	3.24	4200	4.20	416.50	52	334.25	100
7579	3.26	4158	4.22	413.40	53		
7462	3.28	4118	4.24	410.41	54		

CLTS Resistance-to-Temperature Conversions

CLTS temperature sensors are mounted on the magnet to detect the link and shield temperatures. The resistance of a CLTS sensor increases as its temperature rises. Table A-6 lists the temperatures that correspond to various resistances of the CLTS temperature sensors.

Table A-6CLTS Resistance-to-Temperature Conversions

CLTS Sensor Reading (Ω)	Temperature (Kelvin)	CLTS Sensor Reading (Ω)	Temperature (Kelvin)	CLTS Sensor Reading (Ω)	Temperature (Kelvin)	CLTS Sensor Reading (Ω)	Temperature (Kelvin)
220.1	10	231.7	58	243.3	106	267.3	205
220.6	12	232.2	60	243.8	108	268.5	210
221.1	14	232.7	62	244.3	110	269.7	215
221.6	16	233.2	64	244.8	112	270.9	220
222.0	18	233.6	66	245.3	114	272.1	225
222.5	20	234.1	68	245.7	116	273.3	230
223.0	22	234.6	70	246.2	118	274.5	235
223.5	24	235.1	72	246.7	120	275.7	240
224.0	26	235.6	74	247.9	125	276.9	245
224.5	28	236.1	76	249.1	130	278.1	250
224.9	30	236.5	78	250.3	135	279.3	255
225.4	32	237.0	80	251.5	140	280.6	260
225.9	34	237.5	82	252.7	145	281.8	265
226.4	36	238.0	84	254.0	150	283.0	270
226.9	38	238.5	86	255.2	155	282.2	275
227.4	40	239.0	88	256.4	160	285.4	280
227.8	42	239.4	90	257.6	165	286.6	285
228.3	44	239.9	92	258.8	170	287.8	290
228.8	46	240.4	94	260.0	175	289.0	295
229.3	48	240.9	96	261.2	180	290.2	300
229.8	50	241.4	98	262.4	185	291.4	305
230.3	52	241.9	100	263.6	190	292.6	310
230.7	54	242.3	102	264.8	195		
231.2	56	242.8	104	266.0	200		

Note: This table is not valid for Pt100 sensors, including for link and shield on magnets with serial numbers >50.

Appendix B Shipping Box Contents and Installation Tool Kits

Shipping Containers

The following table lists the contents of the MR system shipping containers.

Table B-1 Shipping Box Contents

Box #	Description	Part Number	Quantity
1	Magnet	1015-0000 or 1015-0001	1
1	Gradient coil	1009-0000	1
2	Compressor, model 71A or model 31C	1099-0022 or 1099-0040	1
3	Helium supply and return lines		2
4	Coldhead assembly	2001-0012	1
5	Magnet saddle stand	2001-1000	1
5	Sst. flex tubing, 48" (1219 mm)	1067-0005	1
5	Coldhead support plate	2001-1018	1
5	Miscellaneous enclosure parts	2001-0013	
5	Miscellaneous magnet parts	1015-0001	
5	Pressure transducer	2001-0004	1
5	RF coil interface plate	2001-0014	1
5	Laser light assembly	2001-0005	1
5	Heel support pad	2001-1013	1
5	Heel support lock knob	2001-1016	1
5	Hardware kit	2000-7701	1
6	Magnet front cover	2001-1002	1
6	Magnet rear cover	2001-1003	1
6	Foam wrap	1085-0002	
7	Cylindrical cover	2001-1044	1
7	Heel support platform	2001-1012	1
7	Extremity support bracket	2001-1011	1
7	Step stool	1099-0053	1
7	Teardrop cable cover	2001-1030	1
8	Right enclosure base	2001-1005	1
8	Left enclosure base	2001-1006	1
9	Teardrop enclosure	2001-1004	1
10	patient chair assembly	2006-0001	1
11	180mm RF coil	2003-0000	1
11	123mm RF coil	2003-0001	1
12	Penetration panel	2005-0006	1
12	Label kit	2000-7702	1
13	Electronics cabinet	2005-0012	1
14	LCD computer monitor	1075-0000	1
14	Host computer, keyboard, and mouse	2005-0008	1
14	Power control unit	2005-0001	1
14	Pressure control module	2001-0003	1
14	Phantom kit	2000-0005	1

Table B-1 Shipping Box Contents (Continued)

14	Operator's Guide	2000-9001	1
14	Viewer's Guide	2007-7504	1
14	Software Release Notes	2007-4301	1
14	MR Exclusion Warning Signs	1091-0010	1
14	MRI Health Effects	1099-0030	1
15	System electronics to penetration panel kit	2005-7701	1
15	Exam room cable kit	2005-7702	1
15	Rack to console cable kit	2005-7703	1
15	50-foot (15 m) extension(?) console cable kit	2005-7704	1
16	Penetration panel cover	1099-0044	1
17	Codonics component kit	2000-7705	1

Tool Kits

Transfer Tool Kit

Table B-2 Transfer Kit Contents

Description	Quantity
Wheel case 30"W x 30"L x 10"H (762 x 762 x 254 mm)	
Cryogen transfer line extension, stainless with brass tip 3 ft. (0.9 m)	1
Cryogen transfer line, braided stainless, 6 ft. (1.8 m) S/N C-132-055-5-002	1
Face shield, protective, blue/clear, Cabot Z-87	1
Short transfer line extension (43 mm)	1
Tubing, gum rubber, 10 ft x 5/8 in (3 m x 16 mm)	1
Compressor charging kit	1
Blow-out hose with K-flange	1
3/8" x 1/2" fitting for dewar	1
1/8" X 1/4" fitting for regulator	1
3/8" x 1/4" fitting for regulator	1
Miscellaneous clamps	4
#28 O-ring	3
Magnet vent valve O-Ring	1
Dewar christmas tree	1
Regulator	1
Gloves, leather, pair	1
Total weight of kit (lbs.)	44 (20 kg)
Volume of case (cu. ft.)	3.9 (0.11 m ³)

Magnet Dolly Tool Kit

Table B-3 Magnet Dolly Kit

Description	Quantity
Wheel case 26" W x 38" L x 22" H (660 x 965 x 559 mm)	
Assembly hardware, bolts, nuts, & washers, in plastic bags	32
Jacks, wheel type, Fulton XP15L	4
Dolly end plates, aluminum, 5-1/2" x 32" x 1/2" (140 x 813 x 13 mm)	2
Dolly side plates, aluminum, 13-1/2" x 31" x 1/2" (343 x 787 x 13 mm)	2
Total weight of kit (lbs.)	165 (75 kg)
Volume of case (cu. ft.)	12.6 (0.36 m ³)

Ramp Tool Kit

Table B-4 Ramp Kit Contents

Description	Quantity
Wheel case 26"W x 29"L x 22"H (660 x 737 x 559 mm)	1
Bolt and washer set, brass, 1/4-20 x 3/4 in	3
Cable, high current, 1/0 AWG x 2, 25 ft (7.6 m), with copper end loops	1
Cable, RG58, 20AWG 15 ft (4.6 m), w/ BNC connectors and alligator clip adapters	1
Diode, power rectifier, type IN2054R, with brass bracket	2
Z-Brackets	2
Meter, NMR tesla, Metro Lab Group, 3-PR2025	1
Packing material, grey foam, various sizes and shapes	23
Power cord, black, 5 ft (1.5 m)	2
Power supply, 0 to 30 VDC 3A, Tenma	1
Power supply, 6 volt 200 amp, Xantrex XFR6-200, or AMETEK XG6-220	1
Probe, magnet energizing, stainless with copper connectors S/N QUE114951	1
Probe, sensing, for tesla meter, 1062-4-10m with cable	1
Resistor, 2 Ohms, 20 watts, type HLW-20-AIZ, with leads	1
Screw, nut, & washer set, steel, 1/4-20 x 3/4 in	2
Total weight of kit (lbs.)	111 (50 kg)
Volume of case (cu. ft.)	9.6 (0.27 m ³)

Calibration Fixture Tool Kit

Table B-5 Calibration Fixture Kit Contents

Description	Quantity
Case, 14" W x 24" L x 15" H (356 x 610 x 381 mm)	1
Point sample apparatus P/N 2005-8010	1
Phantom test fixture, Plexiglas cylinder 3-1/4" dia x 4" (83 x 102 mm), grad cal	1
Total weight of kit (lbs.)	8 (3.6 kg)
Volume of case (cu. ft.)	2.9 (0.08 m ³)

Vacuum Pump Tool Kit

Table B-6 Vacuum Pump Kit Contents

Wheel case 26"W x 38"L x 22"H (660 x 965 x 559 mm)	1
Actuator, magnet valve, stainless & brass, 1093-0001	1
Cable, gauge to controller, Pfeiffer PT448 250-T	1
Cable, turbopump to controller, Pfeiffer PM051103-T	1
Cable, RS485	1
Clamps, NW50	3
Clamps, Pfeiffer NW25	1
Clamps, Pfeiffer NW40	5
Clamps, NW16	5
Control, turbopump, PMCO 16948 SN# 76050466	1
Coupler, cross type, NW40 w/NW40 to NW50 mm adapter PF128240-X	1
Fan assembly, Pfeiffer PMZ01253 D35614	1
Filter, oil mist, Pfeiffer ONF016 PKZ40003	1
Gauge, vacuum, full range, Pfeiffer PTR26001, GE SN # 0039	1
Grease, vacuum, silicone, Dow Corning 695400008, tube	1
Manual, instruction, Pfeiffer pump system	1
Oil, vacuum pump, Pfeiffer P3, PK001106-T, 1 Liter	1
O-ring, PF110 216-T	5
O-ring, NW25	1
O-ring, NW50	3
O-rings, NW40	4
Packing material, grey foam, 2 ft x 3 ft x 2.5 in	4
Plate, blank, 16 mm, 40 mm, 50 mm, PF128040	4
Plate, reducer with O-Ring	1
Power cord, orange, 25 ft (7.6 m)	1
Pump, rotary vane vacuum, Pfeiffer PKD41703A, S/N 20137380	1
Pump, turbomolecular vacuum, Pfeiffer PMP02802, S/N 13178376	1
Reducer, NW50-NW40	1
Relay, safety, Pfeiffer PM041938-T D35614	1
Tape, aluminum, adhesive, 1 in (25 mm) wide	1
Tool, all-thread, 5 in x 3/8 in (127 x 9.5 mm)	1
Wrench, open end, ASAHI Tool B4630-8840, Black, #26	1
Wrench, open end, ASAHI Tool B4630-8840, Black, #29	1
Wrench, Super Tool RQ30N, Orange/Black	1
Tubing, stainless, flex, 20 in x 2 in (508 x 51 mm) with K-flanges	1
Tubing, stainless, flex, 40 in x 1 (254 x 25 mm) in with K-flanges	1
Valve, brass, 3000 PSI, B-1KF-2	1
Valve, electro-mechanical, safety, Varian L9940602 with cables	1
Valve, pump-out port, KF25, HPS Div MKS	1
Wrench, spanner, for magnet shipping bolts	1
Total weight of kit (lbs.)	132 (60 kg)
Volume of case (cu. ft.)	12.6 (0.36 m ³)

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